Moodle-Extension for Club and Events

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AALBORG UNIVERSITY
STUDENT REPORT

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Title:

Moodle-Extension for Club and Events

abstract:

Project period:

Autumn 2025

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Number of pages:

117

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Abstraction

This report investigates how fragmented communication of social activities at Aalborg University CPH affects student participation and explores how a centralized digital platform can improve accessibility to information about clubs and events. Through surveys and interviews the project identified that students struggle to maintain an overview of activities because information is spread across multiple platforms such as Moodle, Facebook, Instagram and Discord.

To address this issue, a Moodle-inspired prototype platform was developed to centralize communication related to social activities. The system includes features such as event listings, calendars, club overviews, and event creation tools designed to improve visibility and accessibility. Usability evaluations showed that users were generally able to complete core tasks successfully, indicating that the platform supports the primary user needs identified during the analysis.

The report concludes that a centralized and integrated solution in Moodle has the potential to improve student engagement by reducing communication fragmentation and making social activities easier to access. While the project remains a proof of concept with limitations such as limited mobile responsiveness and lack of full Moodle integration, it demonstrates how a more user-centered communication platform can contribute to a more connected university environment.

Introduction

University life consists of more than academic work. For many students, clubs, committees, and social events are also part of the study environment. At Aalborg University CPH, several social activities and student-run initiatives exist, but information about these opportunities is not always gathered in one place. Instead, students may encounter information through Moodle, social media, posters, or informal communication between other students.

This project investigates how students at Aalborg University CPH experience the current communication of social activities, with a particular focus on student clubs and events. The project takes its starting point in the idea that participation is not only affected by whether students want to take part, but also by whether information about activities is visible, accessible, and easy to understand.

To explore this issue, the group conducted a questionnaire and interviews with students. These methods were used to gain insight into how students currently find information about social activities, what may prevent them from participating, and how the communication of events and clubs could be improved. The findings, which are presented later in the report, suggest that information is often experienced as fragmented across several platforms. This can make it difficult for students to gain a clear overview of available opportunities.

Based on this, the project focuses on lack of information as a possible barrier to social participation. The goal is not to solve all aspects of the university's social environment, but to address one specific challenge: making information about social events and student clubs easier to find and access. Therefore, the project explores how a digital platform can gather key information about social activities in one place and support students in discovering relevant clubs and events.

The proposed solution is developed as a proof of concept for a Moodle-based extension. Since Moodle is already used by students in their daily study life, the solution aims to present information about clubs and events within a familiar context rather

than introducing a completely separate platform. The program is developed and evaluated through a design and implementation process; including requirements prioritization, prototyping, usability testing, and reflection on the final solution.

The report begins by examining social engagement, sense of belonging, and lack of information as relevant themes for the project. Afterwards, the local context at Aalborg University CPH is explored through the group's own questionnaire and interview findings. The report then describes the design process, the implementation of the prototype, and the results of usability testing. Finally, the project is discussed in relation to its limitations, future development possibilities, and overall contribution.

Chapter 1

Research on social engagement and lack of information

1.1 Research on social engagement and lack of information

Before developing a solution, it is necessary to understand the problem area behind the project. This chapter therefore examines social engagement in a university context and explores how students' access to information may influence their participation in clubs, committees, and events.

1.2 Lack of social engagement

Feeling connected to others is an important part of university life[50], but many students struggle with social engagement. Moving to a new environment, balancing coursework, and forming new friends can be challenging, and not everyone finds it easy to build a supportive social network. Research shows that when students are socially isolated, it can affect both their well-being and their academic performance. This section and the section below looks at how loneliness shows up among university students and how it changes over the course of their studies.

Several studies show that loneliness is a widespread issue among university students and can impact both their well-being and academic life. For example research points out that "loneliness is one of the foremost common issues among university students that may negatively affect their physical and mental states and academic performance." This suggests that a lack of social connections can have real consequences for students[59].

Table 1.

The Distribution of Loneliness According to the Demographic Characteristics (n = 538)

Factors	Frequency (n)	Percentage (%)	Loneliness Status		P-value
			Lonely	Not Lonely	
Gender					
Male	237	44.1	56 (23.6)	181 (76.4)	0.116
Female	301	55.9	86 (28.6)	215 (71.4)	
Age (years)					
21 and lower	256	47.6	73 (28.6)	182 (71.4)	0.145
22 and above	282	52.4	69 (24.4)	214 (75.6)	
Marital status					
Single	482	89.6	130(27.0)	352 (73.0)	0.236
Married	56	10.4	12 (21.4)	44 (78.6)	
Residence					
With family	370	68.7	77 (21.9)	274 (78.1)	0.001
Dormitory	168	31.3	65 (34.8)	122 (65.2)	
Year of study					
First	112	20.8	17 (20.2)	67 (79.8)	0.031
Second	256	47.6	38 (33.9)	74 (66.1)	
Third	86	16.0	72 (28.1)	184 (71.9)	
Fourth and above	84	15.6	15 (17.4)	71 (82.6)	

Figure 1.1: Levels of loneliness among students world wide [59].

The study above shows that loneliness affects a large number of students. For example, the table shows that 28.6% of female students and 23.6% of male students reported feeling lonely[59]. First year students report feeling lonely less often (20.2%) than second year (33.9%) and third year students (28.1%)[59]. As new students, they meet many people through orientation and social events, which helps them form friendships. In later years, students have fewer opportunities to meet new people, and levels of loneliness increase. However, fourth year students and above report slightly lower levels (17.4%)[59], possibly because they have established more stable friend groups.

The results suggest that loneliness is something many university students experience at different points during their studies. This shows that building social connections at university is not always easy, but it can play an important role in how students experience their time at university.

1.3 Sense of belonging

This section looks at the concept of belonging and its role in student motivation and academic persistence. Research suggests that when students feel like a valued part of their university community, it has a measurable impact on both their well-being and their likelihood of continuing their studies.

A sense of belonging helps students feel connected and accepted within their campus. When students feel they belong, they are more likely to participate and stay engaged in their studies. Psychological research shows that a student's sense of belonging can have an important impact on their experience at university. When students feel like they are part of the university community, they are often more motivated and more likely to continue their studies.

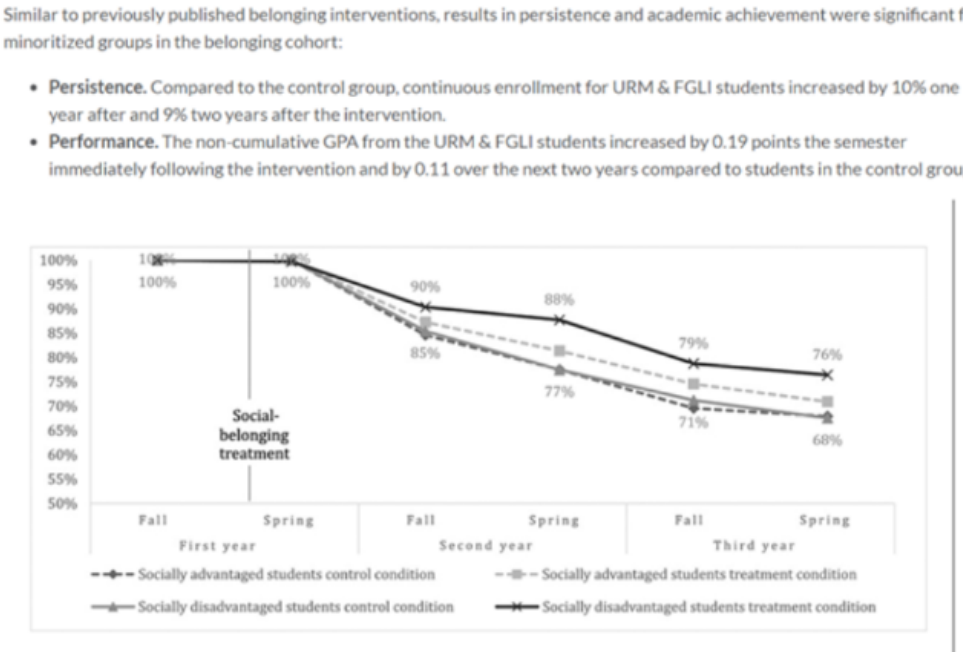


Figure 1.2: Belonging intervention increases continuous enrollment over 2 years by 9 percentage points among socially disadvantaged students enrolled in a broad-access institution [19].

In one study, socially disadvantaged students who took part in a belonging intervention were 10% more likely to stay enrolled after one year and 9% more likely after two years compared to students who did not participate[19]. The study also found small improvements in academic performance, where students' GPA increased by about 0.19 points in the following semester and by 0.11 over the next two years[19]. This shows that by feeling connected to the university environment, it can help students stay engaged and continue their education.

Overall, these findings highlight that social engagement and a sense of belonging play a central role in students' well-being, motivation, and academic success. This is relevant for our case because it shows that challenges related to social participation are not only individual, but also shaped by the university environment.

In the context of Aalborg University CPH, this suggests that improving students' access to social opportunities could strengthen both their sense of belonging and their overall study experience. Therefore, it becomes important to examine what prevents students from engaging in these activities.

To explore this, a survey was conducted among students at the Department of Computer Science at Aalborg University CPH, focusing on possible barriers such as lack of information, lack of appeal, and lack of social network. Based on the responses, lack of information was selected as the primary focus for further investigation.

1.4 Relating existing research to our findings

This section compares existing academic research on student engagement with the data collected from students at Aalborg University CPH. To examine whether similar challenges were present in the local context, both a questionnaire and semi-structured interviews were conducted with students. The questionnaire received 37 responses and was used to gain a broader quantitative understanding of how students engage in social activities and how they perceive the information currently provided about events, clubs, and committees. In addition, eight semi-structured interviews were conducted to gain a deeper qualitative understanding of students' experiences, including how they find information, what challenges they encounter, and what could make participation easier. Appendix A.2 Appendix A.1

Previous research supports the importance of information accessibility in student engagement. A study titled "Blockchain-Based Approach to Foster Student Engagement on Campus" highlights that participation in social and academic activities can benefit students by improving skills, expanding social networks, and enhancing career opportunities. However, the study also identifies lack of awareness and motivation as key barriers to participation [57].

This barrier is directly reflected in the questionnaire data, where students rated the lack of information about social events and committees as an average of 6 out of 10. This indicates that several students perceive access to information as a relevant issue. The interview findings further supported this, as participants described information as being spread across different platforms and communication channels. Some students also explained that they discovered relevant platforms, such as Moodle Students, later in their studies, while others relied on informal communication

through friends or tutors.

Further research shows that age and experience at university can influence participation in campus activities. A study on first-year students' participation found that engagement often develops over time as students become more familiar with the university environment and available opportunities [48]. This is reflected in our interview findings, where newer students appeared more dependent on guidance from others to discover where information about social activities could be found.

Overall, both existing research and the group's own findings indicate that student engagement is influenced not only by motivation, but also by how visible and accessible social opportunities are. When communication is fragmented, unclear, or untimely, students may struggle to access opportunities that could improve their university experience.

This supports the project's focus on lack of information as a central barrier to social participation at Aalborg University CPH.

1.5 Understanding the challenges at Aalborg University CPH

The previous section related existing research on lack of information to the group's own questionnaire and interview findings. While this established lack of information as a relevant issue at Aalborg University CPH, it is necessary to examine the local context in more detail in order to understand how the issue appears in practice.

This section therefore focuses on the students' experiences with finding and accessing information about social activities. First, the qualitative interview data is organized through an affinity diagram in order to identify recurring patterns. Afterwards, a PACT analysis is used to examine the problem from the perspectives of people, activities, contexts, and technologies. Later, the current solutions on Aalborg University CPH is evaluated which afterwards focuses on the more specific solution delivered by the school; Moodle for Students, which is evaluated using selected usability heuristics to understand how well it supports students in accessing information about

clubs and events. Finally, we examine how other universities approach the challenge of encouraging students social engagement, to identify strengths, limitations and possible improvements.

compared to identify strengths, limitations, and possible improvements

1.5.1 Affinity Diagram

Appendix A.3

To make sense of the qualitative data gathered through interviews, an affinity diagram was used as the primary analytical method. This section describes how the diagram was constructed, why this method was chosen over other alternatives, and what themes and patterns emerged from the data. The findings from this analysis form the basis for the subsequent PACT analysis.

To identify patterns and nuances in the qualitative interview data, it was put in an affinity diagram, which is a method used for grouping and organizing findings and more. An affinity diagram was chosen because the interview data consisted of open-ended and diverse responses, making it useful for identifying recurring themes and relationships without imposing predefined categories. Compared to more rigid analytical methods, the affinity diagram supports an exploratory and user-centered approach, allowing patterns to emerge naturally from the participants' statements [25]. This made it particularly suitable for understanding students' experiences and perceptions regarding access to information about social events. The analysis revealed several recurring themes related to how students access and perceive information about social events. When searching for information, students reported relying primarily on social media, public displays such as posters or boards, and verbal communication passed on by friends, while some students indicated that they do not actively search for information at all.

Regarding the quality of information, most participants did not perceive major issues, but some noted that information can be missing, unfinished, or difficult to locate. In terms of coverage, responses indicated that although some information

is sufficient, it can be hard to maintain an overview of events, and in some cases, the information appears superficial or leads to a loss of interest. Experiences from previous events, such as bowling, the study start party, the Christmas lunch, and the Friday bar, were generally associated with clear and satisfactory information, while information about other events was often missing.

Finally, many participants expressed frustration with scattered information across different channels and indicated a preference for having all event-related information gathered in one place, for example in the form of a schedule. This fragmentation of information was also linked to lower motivation to participate and a tendency to forget posts about upcoming events.

Based on these findings, it is necessary to examine the context in which these challenges occur in order to gain a clearer understanding. This is explored through a PACT analysis.

1.5.2 PACT Analysis

Building on the themes identified in the affinity diagram, this section applies the PACT framework to gain a more structured understanding of the challenges students face. the section aims to clarify who is affected by the problem, what they are trying to do, where and how they do it, and what tools they currently rely on.

The PACT framework (People, Activities, Contexts, and Technologies) is used to better understand the factors influencing student engagement with social events. It is a method used of analyzing human-interaction with a product looking at different factors[7],

- People

The primary users are students who want to participate in social activities but differ in how actively they search for information. Some students rely on social media or information shared by friends, while others do not actively look for event information at all. This suggests varying levels of motivation and awareness among students,

where some depend heavily on social communication to discover events.

- Activities

The main activity for students is discovering, understanding, and deciding whether to attend social events. The affinity analysis shows that this process can be difficult because information is sometimes incomplete, hard to find, or scattered across multiple sources. When students cannot easily obtain an overview of events, their interest may decrease, which can reduce participation.

- Context

The context is the university environment where students are exposed to event information in several different places, such as social media platforms, posters, boards, and through conversations with peers. While some larger events provide clear information, smaller or less visible events may lack proper promotion, making it harder for students to discover them.

- Technologies

Students currently rely on multiple technologies and communication channels to access information, including social media and physical displays. However, the affinity analysis indicates that this fragmented distribution of information can lead to confusion and frustration. Several participants expressed a preference for a centralized platform or schedule where all event-related information could be accessed in one place, potentially improving awareness and participation in social activities.

The PACT analysis highlights that students face challenges related to accessing and understanding information due to fragmented communication and varying user needs.

Based on this, the following section evaluates how well the current solutions support students in accessing information.

1.6 State of the art

To establish a foundation for understanding the existing information channels at Aalborg University CPH, this section examines how the university currently provides and communicates information about student organizations, clubs, and events. The purpose is to identify the strengths and limitations of the existing platforms and to assess how well they support students in discovering and engaging with social activities on campus.

1.6.1 Current Solutions at Aalborg University CPH

Before comparing AAU's approach to that of other institutions, this section provides an overview of the platforms students at Aalborg University currently rely on to find information about student organizations, clubs, and events. Understanding the existing products at AAU is necessary in order to evaluate where it falls short and what a more effective solution might look like.

At Aalborg University CPH students currently rely on several platforms to find information about student organizations, clubs and events. One of the main platforms is *Moodle for students*, which provides an overview of various campus clubs.[2] However, many pages redirect students to external platforms such as Instagram or Facebook for further information.

Social media is also widely used by student organizations to share events and updates. While this allows direct communication with students, the information becomes fragmented across multiple pages and groups, making it difficult to gain a complete overview of campus activities.

In addition, information related to student life is available through The Aalborg University CPH website provides .[51] and the *'AAU Student App'*. However, these platforms are mainly focused on study-related information and do not provide a centralized overview of student organizations and social activities.

The Figure 1.3 summarizes the main platforms at Aalborg University CPH, their primary purpose, and their limitations. It demonstrates that while several channels exist, none provide a fully centralized and user-friendly solution for students to discover and engage with campus life.

Platform	Purpose	Limitation
University website	General information about student life	No complete list of all student organizations, information scattered across multiple pages.
Moodle for students	Platform for communities/clubs at campus.	Links redirect to external platforms, difficult to find, require manual enrollment, low awareness among new students, unclear how to contact administrators.
Facebook	Events and student groups	Information is scattered across groups and pages, no central overview, students must request access to join.
Instagram	Updates and promotion of activities	Posts disappear quickly, events are hard to find later, no structured list, mixes of general content and events.
AAU Student App	Schedule, study information, and announcements	Limited overview clubs, not focused on student life, not up to date with events

Figure 1.3: Summary Table of Aalborg University CPH Platforms. The table highlights how fragmented information across current channels creates an obstacle for student engagement.

Since Moodle for Students represents one of the central university-provided platforms for communicating information about clubs and social activities, the following section focuses more specifically on evaluating its usability.

1.6.2 Usability evaluation of the current solution Moodle for students

In this section, the current solution, Moodle for students[1], is evaluated using selected principles from Nielsen's heuristics[36] to assess its usability and to support the key issues highlighted by respondents in the conducted survey and interviews. Appendix A.2

To begin with, conceptual scenarios have been developed based on the PACT analysis and the experiences described by respondents in the aforementioned survey and interviews which covers both students as participants in club events and students as organizers. The method of making conceptual scenarios is to visualize different outcomes of a user interacting with a system[18]. And the exact purpose here is to showcase the interaction-flow between the user and the current solution together with clarifying and highlighting the core functionalities and critical pain points that the current solution either supports or fails to support at present.

In the first two scenarios the actor is an arbitrary student, their motivation is to engage with fellow students and their goal is to participate in a student club activity.

Scenario A:

The student is familiar with the platform Moodle for students and therefore uses it as primary place to search for information about social initiatives.

While navigating the platform, the student is unable to find sufficient information about how to participate in initiatives or clubs.

The platform provides either a link to another platform or contact information related to the activity. The student follows the link or investigates the provided information

However, the student still cannot find the necessary information needed to get started.

To proceed, the students must contact a responsible coordinator to obtain the relevant details.

After contacting the coordinator, the student finally receives the necessary information and is able to participate in the activity.

The students worries if other students will attend.

The student achieves their goal.

Figure 1.4: Scenario A drafted by the group.

Scenario B:

The student is not familiar with the platform Moodle for students.

The student therefore must independently search for places where relevant information might be available. The students ask other students/looks around campus to find information/search online

The student finds various Facebook-groups and other social media sites.

The available information is insufficient and the student must contact a coordinator to obtain relevant details.

The student does not receive a response from the coordinator.

The student ultimately does not find the information needed.

The student gives up and does not achieve their goal.

Figure 1.5: Scenario B drafted by the group.

In the last scenario the actor is an event-coordinator, their motivation is to facilitate social engagement and their goal is to make an event that brings as many students together as possible.

Scenario C:

The organizer is familiar with the platform Moodle for students.

The organizer won't use the platform to promote their event due to not being able to reach people.

The organizer creates an event and promotes it on several So-Me platforms.

The organizer sends a message on several Messenger groups and to individual people from other semesters in order to reach most people.

The organizer reaches only a few people and cancels the event.

The organizer does not achieve their goal.

Figure 1.6: Scenario C drafted by the group.

Based on these scenarios, it becomes evident that the current solution, among other things, conflicts with Nielsen's seventh heuristic, *flexibility and efficiency of use*[36]. The different actors have varying starting points, which either enable or limit their ability to make use of the current solution. For instance, users who are familiar with

the concept of course-enrollment and who are active on social media have a greater likelihood of achieving their goal.

This suggests that the existing platform where information about student committees and clubs is available is not clearly communicated. And respondents also express that they have been unaware of the platform's existence and that this kind of information is centralized and stored somewhere available to them.

If a student already possesses, or has acquired, the necessary experience and familiarity with the system, Moodle for Students, the structure and format become recognizable and is standardized. The platform is organized similarly to the way the students' courses are structured. As highlighted in Nielsen's second heuristic, *match between the system and the real world*[36], this makes the platform easier for students to navigate, as it builds upon an interface and concept, they are already familiar with and which align with their mental model; their prediction and expectation of the specific system[10].

However, several students report that they are not aware of this feature or even of the platform itself. As a result, students must independently discover that the solution exists and determine how they can access or become part of it. This conflicts with Nielsen's sixth heuristic, *recognition rather than recall*[36], which emphasizes that users should not be required to remember that certain solutions or opportunities exist. Instead, systems should make relevant options visible and easily recognizable for the users.

On the Moodle for Students page, a short description introduces the platform that gathers the university's committees and clubs. The description states:

> "This site has been created for you to have one place for all the information you need about social activities on campus and opportunities that you can take advantage of."[1]

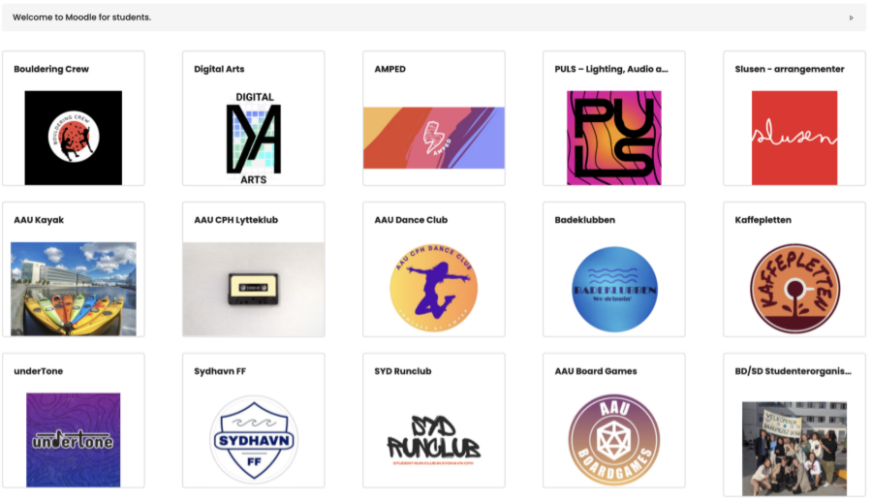


Figure 1.7: Moodle for students interface.[1] This site requires access with login.

It is stated that the page has been created to collect all information about social activities in one place. The different clubnames and their logos are displayed and can be interacted with. Each club includes a description, as well as some form of contact information, and additional links to external pages where further information may be obtained if the provided information is not sufficient.

However, the introductory description becomes somewhat misleading. While the student may expect to find all information gathered in one place, they instead experience that the information is spread across several platforms. This forces the student to make themselves familiar with several systems because the various clubs and committees rely on different and individual solutions for communicating information. This is a pain point to all the actors in all scenarios. It contradicts Nielsen's fourth and fifth heuristics, which emphasize the importance of *consistency* and *standardized*[36] design patterns for clarity, as well as *error prevention*[36] to avoid misunderstandings.

This issue is also a recurring point of criticism expressed by respondents, who describe the experience as "*confusing*" and "*frustrating*," and in some cases as a reason for not engaging in the social life on campus at all.

The evaluation of the current solution using scenarios and selected usability heuristics, reveals several inconsistencies with general usability principles. These inconsistencies may contribute to situations in which the users fail to achieve their intended goals, suggesting that the current solution is not sufficiently user-friendly.

The users, from an organizers perspective; must work on several platforms in order to communicate their information, and from an attendants perspective; they must actively seek out the sources where the information is communicated and subsequently search for the specific information they need. This creates a challenge in the interaction-flow and in the transition between systems. In practice, the user journey often follows a sequence such as:

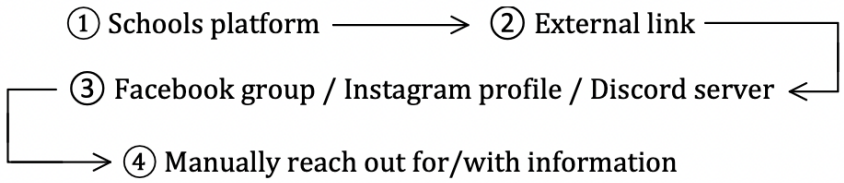


Figure 1.8: Interaction flow showing the transition between official school platforms and external social media channels (drafted by the group).

The lack of usability and the insufficient communication of information results in students missing relevant information and therefore consciously or unconsciously staying out from events and social initiatives. In other words, it limits the participation and reduces the social engagement on campus.

It should, however, be noted that this issue does not necessarily stem from the individual systems themselves. Rather, the problem emerges from the way these systems function together as a combined solution, which in its current form does not provide an optimal user experience.

Since the current structure does not provide students with a clear and centralized overview, it becomes relevant to examine how similar challenges are handled at other universities. This comparison can help identify alternative approaches and design practices that can contribute to the development of a

more integrated solution.

1.7 Comparison with Other Universities

Having described the current solutions at Aalborg University, this section broadens the perspective by examining how other universities handle the communication of student organizations, clubs, and events. The purpose of this comparison is to identify whether more effective solutions already exist and to draw inspiration for improving the current approach at AAU.

To place Aalborg University CPH's current solution in a broader context, this section examines how selected universities present information about student organizations, clubs, and events. The comparison focuses primarily on Danish universities, while also including selected international examples to identify more centralized and structured approaches to communicating student life information.

The purpose of the comparison is to identify whether other institutions provide solutions that address some of the limitations found in Aalborg University CPH's current setup. In particular, the comparison focuses on whether information is easy to find, gathered in one place, and presented in a way that supports students in discovering relevant organizations and activities.

To structure the comparison, a set of evaluation criteria was defined based on the issues identified in the previous analysis. These include whether the information is centralized or fragmented, whether there is a clear list of organizations, whether a central event calendar exists, whether organizations are described with relevant information, how easy the navigation is, and whether there is a dedicated student life page.

Figure 1.9 provides an overview of how the analyzed universities perform according to these criteria. The comparison shows that most universities provide some information about student organizations and events, but that the struc-

ture, accessibility, and level of centralization vary significantly.

University	Not Fragmented	Clear Organization List	Event Calendar	Organizations Explained	Easy Navigation	Central Student Life Page
IT University of Copenhagen	Yes	Yes	Partial	Yes	Easy	Yes
Copenhagen Business School	Partial	No	Partial	Partial	Medium	Partial
Technical University of Denmark	No	Yes	Partial	Yes	Hard	Partial
University of Southern Denmark	No	Partial	Partial	Partial	Hard	No
Aarhus University	No	No	Partial	Partial	Medium	Partial
Aalborg Universitet	No	Yes	No	Partial	Medium-Hard	Partial
Massachusetts Institute of Technology	Yes	Yes	No	Yes	Easy	Yes

Figure 1.9: Comparison of university platforms for presenting student organizations and events.

Among the universities analyzed, the IT University of Copenhagen stands out as one of the most centralized and accessible examples. ITU provides a central platform where students can access information about events, clubs, and organizations in one place.[51] The platform includes a clear list of organizations, descriptions of each club, a calendar of activities, and visible contact information. Because the information is centralized and well structured, students can more easily gain an overview of available opportunities. For this reason, the ITU solution serves as a relevant example of how student life information can be organized more clearly.

Beyond Danish universities, international institutions also demonstrate centralized approaches to presenting student organizations and activities. A notable example is the Massachusetts Institute of Technology (MIT), which provides a page called "Clubs and Activities" that introduces student organizations and links to a platform where students can access a comprehensive overview of registered organizations.[28] The platform provides a searchable

database containing the full list of clubs at the university, allowing students to filter organizations by categories and interests. Each organization includes a description and relevant contact information, making it easier for students to understand the purpose of the club and how to join. This structured system provides a clear and centralized overview of campus activities and reduces the reliance on external platforms or informal communication channels.

This is relevant not only from a usability perspective, but also in relation to student engagement more broadly. Research from MIT highlights the importance of student engagement in extracurricular activities. Surveys conducted within MIT's recreational and club sports programs show that students who participate in organized activities report lower levels of isolation and improved social well-being compared to the general student population.[28] This suggests that accessible and well-organized information about student organizations can play an important role in encouraging participation and strengthening the social environment at universities.

In contrast, the other universities analyzed, including Copenhagen Business School, the Technical University of Denmark, the University of Southern Denmark, Aarhus University, and Aalborg University CPH, generally provide less centralized solutions. Although information about student life exists, it is often scattered across multiple pages or platforms. In several cases, students must navigate through multiple layers of pages before finding relevant information, making it harder to discover organizations and events.

Overall, the comparison shows that while universities recognize the importance of student organizations and campus activities, many current solutions remain fragmented and difficult to navigate. This reinforces the findings from the analysis of Aalborg University CPH and highlights the need for a more centralized and user-friendly platform that allows students to easily explore and engage with campus life.

1.8 Problem statement

This section brings the previous findings together into a problem statement that defines the core challenge the project aims to address, and that will guide the design and development work in the following chapters.

The problem analysis shows that social engagement plays an important role in students' well-being and academic experience, additionally it highlights that lack of accessible information can act as a barrier to participation. Also it has become evident students at Aalborg University face significant challenges when trying to access information about student organizations, clubs, and events.

Although there are several digital platforms available, such as Moodle for students, social media, the university website, and the 'AAU Student App', these solutions function independently rather than as a coherent system. As a result, students must navigate multiple platforms to gain an overview of campus life. This fragmented structure disrupts the flow of interactions, increases cognitive load, and creates confusion.

The usability evaluation supports this conclusion. The current combined solution conflicts with Nielsen's heuristics, particularly consistency and standardized design. When students repeatedly shift between systems with different structures and communication formats, the risk of misunderstandings and missed information increases. Importantly, the core issue does not lie within the individual platforms themselves, but in the lack of integration between them. Together, they fail to provide a centralized and intuitive user experience.

Student feedback further emphasizes the consequences of this fragmentation. Many describe the process as frustrating and confusing, and some indicate that the difficulty of finding reliable information discourages them from participating in social activities. Because students must actively search across platforms and often rely on informal communication, relevant events and opportunities are easily overlooked. This limits participation and may negatively affect social

engagement on campus.

The comparison with other universities reinforces the need for improvement. For example, the IT University of Copenhagen offers a centralized and structured overview of clubs and events, while the Massachusetts Institute of Technology provides a searchable database of student organizations. These examples demonstrate that integrated and user-friendly solutions are achievable and effective in supporting student involvement.

Overall, the analysis indicates that Aalborg University CPH's current communication structure creates unnecessary barriers to social participation. Therefore, the chosen problem statement:

"How can we develop a digital platform that consolidates key information on social events and student clubs at Aalborg University CPH, thereby improving accessibility and encouraging student participation and engagement?"

is well justified, as it directly addresses the identified usability challenges and their impact on student engagement.

Based on the identified challenges, the focus now shifts towards developing a solution. To address the issues related to fragmented and inaccessible information, a set of features has been defined.

Chapter 2

The design process

This chapter describes the process of designing a solution that addresses the identified challenges. The chapter begins by prioritizing the features of the platform using the MoSCoW method. It then presents the prototyping process, followed by a usability test that evaluated whether the design met user needs and expectations.

2.1 MoSCoW prioritization

This section applies the MoSCoW method to categorize the planned functionality of the platform into four groups: Must have, Should have, Could have, and Won't have. The prioritization is based on insights from the survey and interviews, as well as the findings from the problem analysis, and ensures that development remains focused on the features that matter most to users.

The MoSCoW method is used to prioritize system requirements by categorizing them into Must have, Should have, and Won't have[4]. This approach ensures that the most essential features are developed first, while less critical functionalities are considered if time and resources allow.

The following sections present the prioritized features for this project for both

regular students and club owners. Each category includes a justification based on insights gathered from our survey and questionnaire, as well as findings from existing research on student well-being, social engagement, and communication challenges. This ensures that the prioritization reflects both user needs and documented issues within the university social environment. Appendix A.2 Appendix A.1

2.1.1 Must have

The "Must have" requirements include the essential core functionalities that the system cannot function without. These features form the foundation of the platform and ensures that the primary purpose of the system is fulfilled. Without these requirements, the platform would not be able to operate or deliver its intended value to users. Therefore, all "Must have" features are implemented in the initial version of the system.

Club list	A user must be able to view a list of all clubs in one place	Students experience that information is spread across multiple platforms, which makes it difficult to get an overview. For example, one student stated, "Foretrækker et samlet sted med al information" (Person 1). This highlights the importance of gathering all clubs in one place to improve accessibility and navigation.
Information page about event	A user must be able to access a detailed information page for each event	Even though students often receive basic information, they lack deeper insight into events, which affects their motivation to participate. This is supported by a participant who mentioned, "Mangler ofte dybere beskrivelser" (Person 1). A detailed event page ensures that users have access to all relevant information.
Coming events	A user must be able to see an overview of upcoming events	Some students do not actively search for events because it requires too much effort when information is not centralized. One participant explained, "Opsøger ikke, da det ikke er samlet et sted" (Person 3). Providing an overview of upcoming events reduces this barrier and increases visibility.
Creating events + template (Club owner)	A user must be able to create and publish events within their own club, using a structured template.	It is necessary for club owners to be able to create and publish events within their own club, as this is the foundation for having events on the platform at all. Without this functionality, there would be no events for students to view or participate in. An event template helps guide the creation of events.

Table 2.1: Table of "Must have" requirements.

2.1.2 Should have

The "Should have" requirements include features that add significant value to the platform by improving usability, engagement, and overall user experience.

However, they are not essential for the core functionality of the system and can therefore be implemented after the “Must have” features without preventing the platform from working.

Apply to create your own club/event (Regular student)	A user should be able to apply to create their own club or event through a form.	The ability to apply to create their own club or event through a form supports inclusivity and encourages active participation. Despite this, it is not critical in the early stages, as the platform can still function with events created only by selected users. This makes it a valuable but non-essential feature.
Sign up for events (Regular student)	A user should be able to sign up for events.	The option to sign up for events increases user engagement by transforming the platform from an informational space into an interactive one. It allows students to actively commit to events and creates a clearer connection between users and activities. Despite this, the core purpose of the platform, which is to inform students about available events, can still be fulfilled without a sign-up feature.
Display number of attendants (Both parties)	A user should be able to see how many people are attending an event.	The display of the number of attendants was mentioned as an important factor during our interviews. Users expressed that knowing how many people are attending helps them assess the scale and social aspect of an event, which can influence their decision to participate. However, this feature is not required for the platform to operate, as users can still view and attend events without this information.

Table 2.2: Table of the ”Should have” requirements.

2.1.3 Could have

The ”Could have” requirements include additional features that provide extra functionality and enhance the overall user experience. These features are not necessary for the system and are considered lower priority. They may be implemented if time and resources allow, but their absence does not impact the core opperation of the platform.

Admin features	Admin has control over events/clubs that are created	Admin control is not required for MVP functionality. The system can operate without a moderation layer in the early version and add complexity without directly improving student engagement.
Easy sign-up shortcut/button	A short cut that made it so you can sign up from the calendar	This is not a feature that is key to our platform and thus more of an afterthought, that would make it an easier task to sign up to repeat events from clubs.
Contact	Contacting club owners and other students though our platform	We decided not to include the option to contact other students; this is because students can already contact each other using other means.
Filters	Filtering for different fields of study, so that users can easily sort between other students	Filters would help navigation when volume grows (more clubs/events). Furthermore, it would improve relevance of content shown however, this is not a top priority because it does not directly impact student engagement.
Undo	Undo button for applying to events	It is not critical to system functionality, and it does not contribute directly to core goals (discovery and participation).
Notification	A notification that reminds you of upcoming events	This function is not necessary for our platform as it does not further our goal. It will not have a big impact on our product, but it would be nice to include it.
Be part of the club	The ability to join clubs as a member and receive updates	A membership system is not required for event discovery, therefore club membership is not required when clubs still function as event providers.

Table 2.3: Table of the "Could have" requirements.

2.1.4 Won't have

Our product wont include messaging and other social features. Messaging is already an incorporated part of Moodle, as our webpage is based on a concept and meant to be an extension of the Moodle application/website. It's not a feature we are going to develop, but including easy access to message a club/event owner would be something our extension could include if it were to be integrated into Moodle. The need to message someone directly on Moodle is also not a high priority, as students still have the ability to message someone

over their email or phone number, which is linked to a club owner when they fill out their club owner form.

Overall, the prioritization of requirements using the MoSCoW method ensures that the system focuses on delivering a functional and user-centered minimum viable product. This structure prioritization allows the development to remain focused on essential functionality while still leaving room for future expansion based on user feedback and further research.

2.2 The prototyping process

This section presents the considerations behind the developed prototypes. The prototyping process is based on the previously developed MoSCoW model, in which functionalities were prioritized according to user needs and importance. In addition, the process was guided by a design approach in based on the distinction between horizontal and vertical prototyping, which was presented in the course teaching materials about the process of User Centered Design [15]. In this project, the focus has primarily been on vertical prototyping, an approach and method where the most essential functions are developed first and in greater detail. This differs from horizontal prototyping, where a broader range of functions is developed simultaneously, but with less depth [15]. This prioritization was chosen to prevent crucial key functions from being overlooked and to ensure that they work as intended.

2.2.1 Lo-fi prototypes

The design process began with the group considering the product requirements, and then individually making lo-fi prototypes in the form of paper sketches as presented in Figure 2.1. Although not realistic, these sketches made it possible to quickly test ideas, adjust, and further develop layouts and functions without significant costs[43]. After presenting and discussing individual sketches and

ideas, the group agreed on a shared foundation that combined the strongest elements from each proposal.

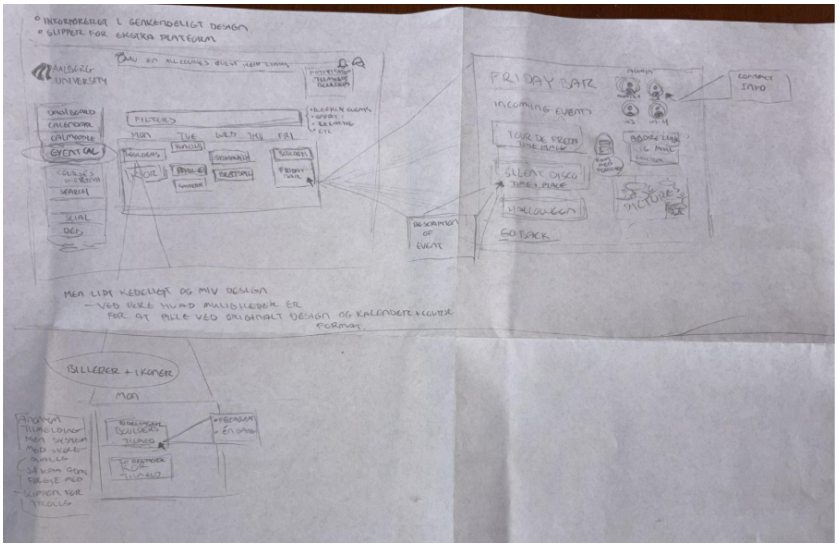


Figure 2.1: Example of lo-fi prototype papersketch of Moodle extension.

2.2.2 Hi-fi prototypes and further design choices

Based on the selected concepts from the lo-fi phase, the prototypes were subsequently developed into hi-fi prototypes using Figma, where the visual design, navigation structure, and user experience is be presented in greater detail and with interactive features[43].

As previously described in section 1.6.2 , the interaction flow constitutes a central challenge, as users currently need to navigate between several different platforms in order to achieve their goals. This regards both the regular student who wishes to participate in events and clubs, and the organizers who need to communicate their events and reach potential participants.

A central design choice has therefore been to develop the solution as an extension of the existing learning platform Moodle, reflecting heuristics such as *consistency & standards* and *recognition rather than recall*[36]. This decision was made with the aim of creating familiarity for the user, as the structure and design of the platform are already part of the student's daily digital habits,

while also ensuring equal opportunities for everyone to participate.

Additionally focus has been on making social activities visible directly within the user's existing flow, so that opportunities for social participation and engagement are presented naturally, without requiring active searching or navigation through external websites. This is achieved through an extension of the already existing left-sided sidebar, which is presented in Figure 2.2, and which functions as a navigation-menu, giving users direct access to social features, when roaming around on the cite.

It should be highlighted that our solution is developed as a proof of concept that demonstrates how the lack of information about social activities, clubs, and committees at Aalborg University CPH can be improved through a more centralized platform. The development does not operate directly within Moodle, Aalborg University CPH's online learning platform, but has been created as a prototype to showcase the functionality and potential of the idea. The solution is not intended to be a separate website, but an extension of Moodle.

Another central design choice has been to make a quick and easy overview of the upcoming events. Entering the page, the user meets a calendar displaying upcoming events for the current week which is presented in Figure 2.2.

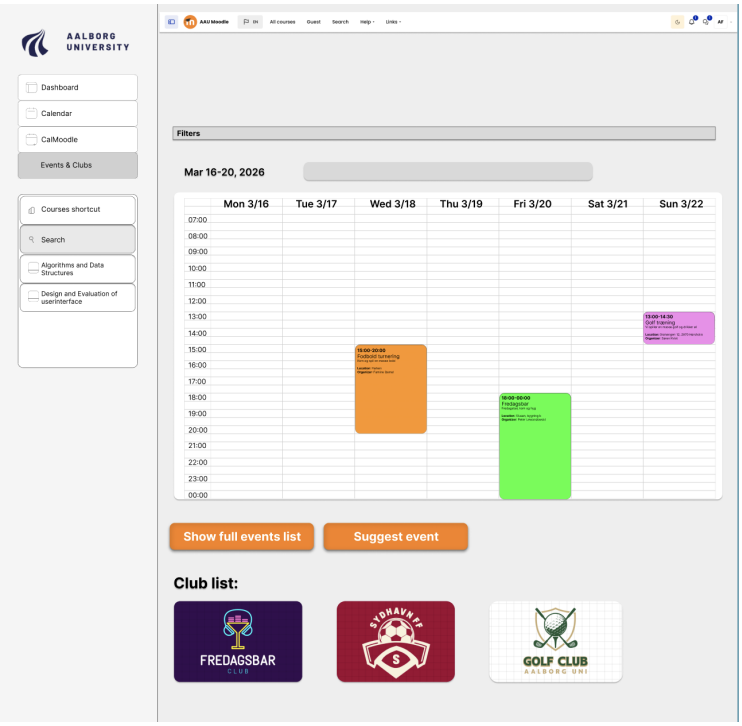


Figure 2.2: Example of hi-fi prototype of Moodle extension homepage made with Figma.

From here, the user can navigate to information pages, which is presented in Figure 2.3, containing relevant details about each event. Two complete lists of both clubs and upcoming events are also available, providing users with a broader overview of where they can access information about the clubs and their upcoming activities.

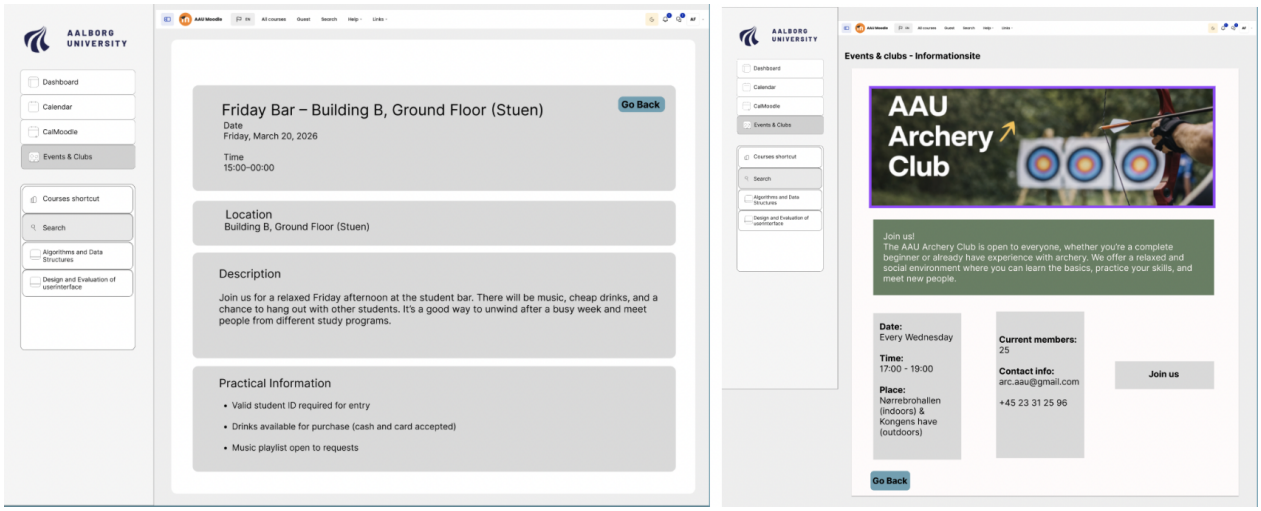


Figure 2.3: Example of hi-fi prototype - Informationpages made with Figma.

Furthermore it is crucial that organizers are able to create events efficiently while ensuring that all relevant information is provided. Therefore, a standardized template must be completed before an event can be published. This is intended to ensure clarity and consistent communication, and is presented i Figure 2.4

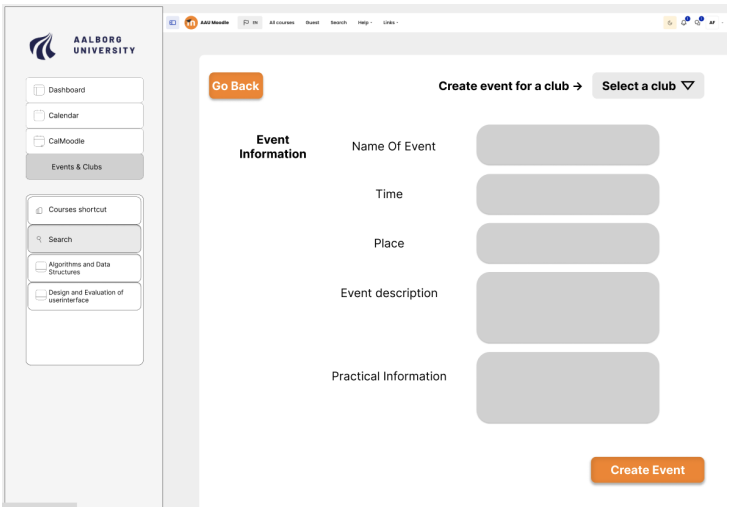


Figure 2.4: Example of hi-fi prototype - Create event template made with Figma.

In addition to the presented core functionalities, several supplementary features were considered and partially prototyped. However, as these were cat-

egorized as Should Have rather than Must Have in the MoSCoW model, the primary focus remained on the essential functionalities presented above. By prioritizing these core functions, the solution aims to make information easily accessible and comprehensive, thereby reducing the need for users to rely on external platforms or other sources for necessary information.

Overall, the prototyping process transformed identified user needs into a functional and visually detailed solution. The resulting hi-fi prototypes incorporate the prioritized core functionalities and serve as the basis for the subsequent usability testing.

To ensure that these solutions align with user needs, it is essential to involve users in the design process. As designers are not the end users, the prototypes are evaluated through usability testing[30] to validate the design decisions and identify potential improvements.

2.3 Usability test on Figma prototype

This section explains how the Figma prototype was tested and presents the findings of the evaluation. These were the key features that needed testing and reviewing: A list of clubs, a calendar overview, an information page about events and clubs, upcoming events, and a template for creating an event. These features were prioritized because they directly address the problem identified in the problem analysis: students have difficulty finding accessible information about social activities. By gathering this information on one platform, the prototype aims to improve visibility, accessibility, and user engagement.

To evaluate the usability of these features, a Think Aloud test[35] was conducted with five participants within the target group. All participants were second semester Software students at Aalborg University CPH. Each test session was conducted individually and facilitated by one group member, while another group member acted as an observer and documented user behavior, comments, and usability issues throughout the session. Each participant was

asked to complete a series of tasks designed to reflect realistic user actions:

- Log in to the platform as a student.
- Find a club you want to join and view its details.
- Create a new club or event.
- Find the number of attendees for the next event you are attending.
- Find an event you would like to attend and register your attendance.

Each participant was asked to continuously verbalize their thoughts while interacting with the prototype. The purpose of the usability test was to evaluate whether the features implemented were intuitive and easy to use, as well as to identify potential areas for improvement.

Choice of Method

The Think Aloud method was specifically chosen for this usability test because it provides insight into the users' thoughts and decision making processes. While other common testing methods such as quantitative surveys or silent observations exist, we chose a more verbalized test. Quantitative surveys and questionnaires only provide statistical numbers and scores, such as a usability rating from 1 to 5. They would show that a student had difficulty finding a club, but they would not explain why or what was confusing about the user interface. Similarly, observing users in silence would show where they clicked, but it would not reveal their internal decision-making process or thoughts. Due to this, we thought that the Think Aloud method would be the best method to use. By forcing users to constantly verbalize their actions, this method provided qualitative insights needed to identify design improvements.

2.3.1 Analysis of Test Results and Application of Feedback

This section presents the findings from the usability test conducted on the Figma prototype. The results are analyzed to identify recurring patterns in

user behavior, both in terms of what worked well and where participants encountered difficulties. The feedback is then used to inform design improvements that will be carried forward into the implementation phase.

Overall, the participants were able to complete the assigned tasks, which indicates that the core Must have functionalities are present and usable. This supports the goal of the prototype, which was to create a minimum viable solution focusing on essential features such as viewing events, joining events, and exploring clubs.

All users were generally successful in finding and joining events. Several participants naturally navigated to the calendar or event list and were able to access relevant information without major difficulties. This suggests that the core structure of the platform aligns well with user expectations. At the same time, differences in user behavior were observed. For example, one participant preferred using the event list rather than the calendar, indicating that multiple ways of accessing information (a Should have feature) can improve usability and should be maintained.

However, the participants also revealed recurring usability issues, particularly related to layout and navigation. Multiple participants expressed confusion regarding the placement of buttons and menu items. For instance, users expected navigation elements such as “Events” and “Clubs” to be more clearly visible at the top of the interface. Similarly, several users expected primary action buttons like “Join” or “Submit” to be placed consistently, often suggesting placement on the right side or bottom of the page. This indicates a mismatch between the system design and common user interface conventions, which can be improved in future iterations.

Another recurring theme was the need for clearer system feedback and visibility. Some users were unsure whether they had successfully joined a club or event, and one participant specifically requested a feature to see their memberships. While this is not part of the core Must have requirements, it represents a valuable Should have feature that can improve user confidence and system

transparency.

The evaluation also highlighted missing or incomplete information in certain areas. For example, users requested more detailed descriptions of clubs, including frequency of activities, as well as access to contact information. This aligns with earlier findings in the report regarding lack of information as a barrier to participation. Although the prototype includes basic information (a Must have), these findings suggest that expanding content and adding contact options (Should have) would significantly improve the user experience.

In addition, some participants experienced difficulty locating specific features, such as creating or suggesting an event. This suggests that while the functionality exists, its discoverability can be improved. Suggested improvements include clearer labeling, better button placement, and the addition of hover text to explain functionality. These are relatively small design changes but can have a significant impact on usability.

Importantly, the feedback must be interpreted in the context of the prototype's scope. Since the prototype was intentionally limited to Must have and selected Should have features, some of the identified issues relate to features that were not prioritized in this iteration. For example, requests for enhanced communication features, better overview of memberships, and more advanced interaction options fall outside the core functionality but provide clear direction for future development.

Overall, the usability test confirms that the prototype successfully supports its primary purpose: providing a centralized platform for discovering and joining events and clubs. At the same time, the feedback highlights several usability improvements, particularly related to layout consistency, navigation clarity, and additional supportive features. These insights will be used to refine the design in future iterations, with a focus on implementing relevant Should have features and improving the overall user experience without compromising the simplicity of the core system.

The Figma prototype now serves as the foundation for the implementation. However, parts of the final design might deviate from the Figma prototype, as relevant adjustments and improvements were highlighted in user feedback from conducted usability tests. Additionally, the design has to be adapted to fit a more Moodle interface and visual style, in order to make the program appear as a natural extension of Moodle

Chapter 3

Implementation

With the design validated through usability testing, this chapter describes how the platform was built. It begins by presenting the technologies and architectural choices that underpin the system, before walking through the frontend and backend in detail. The chapter also covers the usability test conducted on the final implemented program, providing an evaluation of whether the developed solution successfully meets the needs identified during the problem analysis and design process.

Overview

The platform is structured as a traditional multi-page web application, separating frontend and backend responsibilities. On the frontend, HTML, CSS, and JavaScript are used without external UI frameworks, keeping the codebase transparent and straightforward. On the backend, Node.js with Express.js handles routing, server-side logic, and session management, while a Supabase-hosted PostgreSQL database stores all persistent data, including users, clubs, events, and membership records. Club-related images are stored using Supabase's built-in storage solution.

Authentication is kept intentionally simple. Each user is assigned a unique identifier and a role, either Student or Club Owner, which are stored in the database and maintained across pages using Express session management. This

approach reflects the intended use case, as the platform is designed to extend Moodle, where users would already be authenticated before accessing the system.

The implementation is divided into two main areas: the frontend and the backend. The frontend section3.1 covers each page of the application, including the Login page, Calendar3.2.1, Club List3.2.2, Create Event form3.2.3, Event List3.2.4, and Club Information page3.2.5. For each page, the data flow and user interactions are described, and a corresponding process flow diagram illustrates how the system responds to user actions. The backend3.3 section covers the server structure, route handling, and database schema that support this functionality. The chapter concludes with an overview of the unit tests written to validate the core backend functions3.3.

Flow Diagrams: System at a Glance

The following diagrams provide an overview of the five main processes in the application, each of which is examined in detail in the sections that follow.

Login 3.2: The user selects a role, either Student or Club Owner. A login request is sent to the server, a session is created, and the user is redirected to the appropriate dashboard, as illustrated in 3.1.

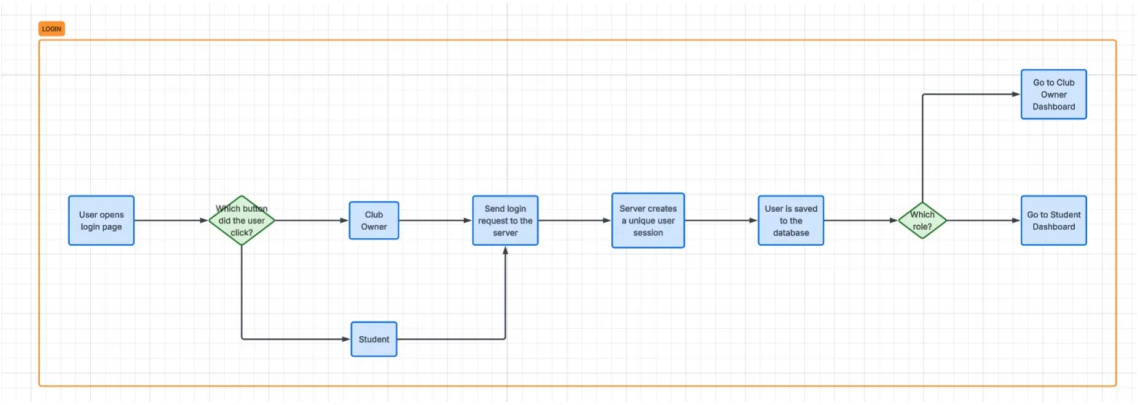


Figure 3.1: Flowchart for the login.

Club List3.2: When the Club List page is loaded, all clubs are retrieved from

the database and displayed as cards. The user's role is read from the session. Clubs can be filtered on the client side, and clicking a card navigates to the corresponding Student or Owner view, as illustrated in 3.2.

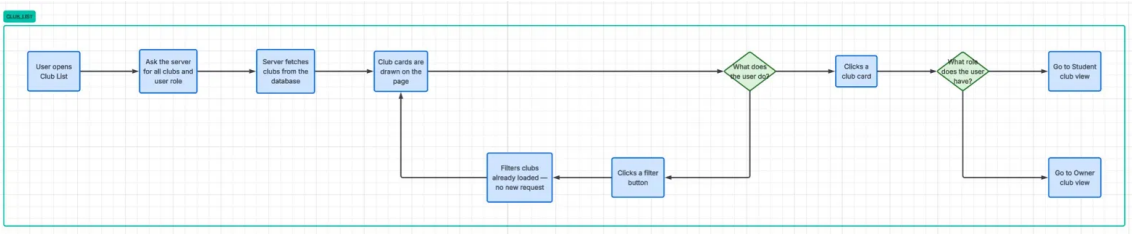


Figure 3.2: Flowchart for the club list.

Club Information Page3.3: Depending on the role stored in the session, the Club Information page either allows a Club Owner to edit club details with an optional image upload or allows a Student to join the club. Membership is saved to the database, and the button state is updated accordingly, as illustrated in 3.3.

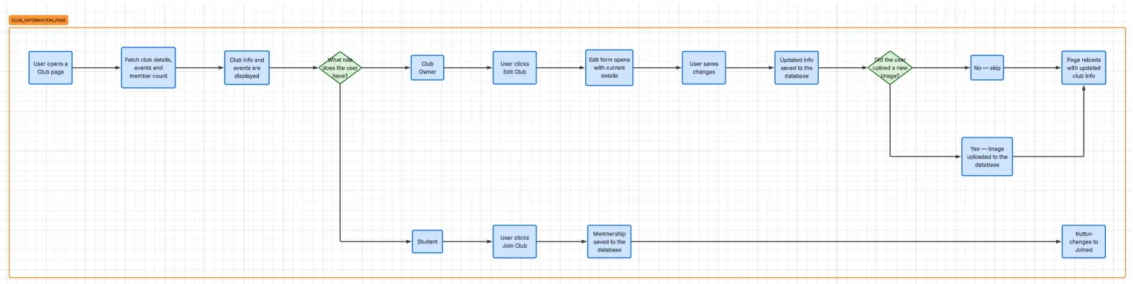


Figure 3.3: Flowchart for the club information page.

Create Event3.2.3: Club Owners are able to create a new event by completing a form that includes a dropdown for selecting a club. Upon submission, all fields are validated for completion. If all required fields are filled, the event data, including a combined start and end time string, is saved to the database and displayed on the page immediately, as illustrated in 3.4.

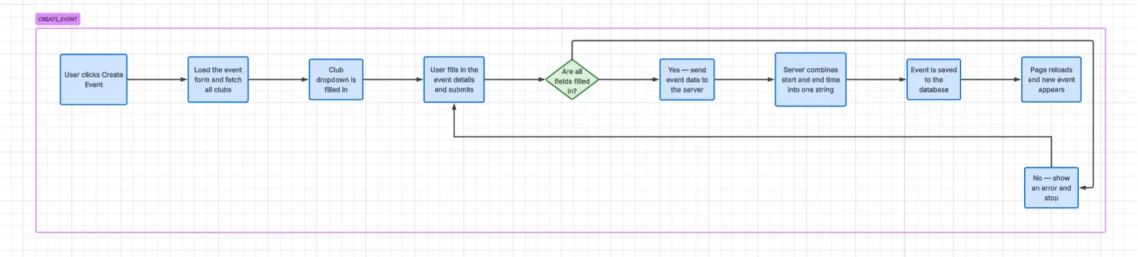


Figure 3.4: Flowchart for creating an event.

Calendar3.5: All events are retrieved from the database and displayed on a weekly grid alongside the user's role. Students are able to switch between weeks and join or leave events directly from the calendar view. Club Owners are able to view the number of members joined to each event through a detailed pop-up, as illustrated in 3.5.

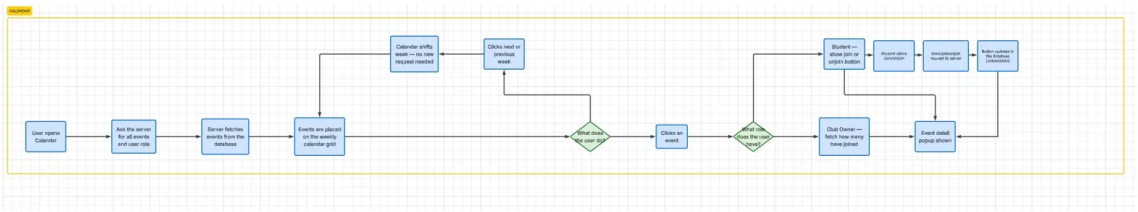


Figure 3.5: Flowchart for the Calendar.

3.1 Technologies

The following section presents the technologies and architectural choices used in the development of the application. The system is built using a traditional multi-page web architecture with a clear separation between frontend and backend responsibilities. The chosen technologies were selected on the basis of simplicity, educational relevance, and suitability for the scope of the project. The section describes the frontend technologies used for structure, styling, and interactivity, as well as the backend technologies responsible for server-side logic, database communication, and session management.

Frontend Technologies

The frontend of the application is built using standard web technologies: HTML, CSS, and JavaScript without the use of any external frontend frameworks such as React or Vue. This approach was chosen due to the alignment of the course content and to maintain simplicity while ensuring a clear understanding and control over the structure, styling, and functionality of the application.

HTML is used to structure all pages of the website, including login page, club listings, and individual club views. Since the application is multi-page, each page is built as a separate HTML document. This makes the structure easy to understand and ensures clear separation between different parts of the application.

CSS is used to handle the visual styling of the application. A custom CSS approach was chosen instead of a framework like Bootstrap or Tailwind, allowing full control over the design and layout. The design has been developed with basic responsiveness in mind. However, the application is not fully optimized for all screen sizes, and its layout is primarily intended for desktop usage. As a result, the responsiveness is limited rather than fully adaptive across all devices.

JavaScript is used to add interactivity and dynamic functionality to the frontend. It handles tasks such as user interactions, form submissions, and communication with the backend server. The Fetch API is used to send and receive data between the frontend and backend, allowing the application to dynamically update content without requiring full page reloads.

Since no frontend framework or additional libraries are used, all logic is implemented using JavaScript. This ensures full transparency in how the application behaves and makes the codebase easier to understand for educational purposes.

Backend Technologies

The backend of the application is built using Node.js with Express.js, which serves as the main server framework. The backend is responsible for handling routing, rendering pages, managing user sessions, and communicating with the database hosted on Supabase.

Express.js is used to create the server and manage HTTP requests and responses. It handles routing between different pages and endpoints, such as login, club data retrieval, and data submissions.[11] Express also renders the HTML pages that make up the frontend, meaning the application follows a traditional multi-page architecture.

Node.js provides the runtime environment for the backend. It allows JavaScript to be executed on the server side, making it possible to build frontend and backend using the same programming language.[39]

Supabase were the chosen database instead of a local database or a self-hosted SQL setup. This decision was made in order to reflect common practices in modern application development, where systems are typically built using external, cloud-based database services instead of locally hosted storage solutions.[54]

Supabase was chosen in particular because it provides a fully managed PostgreSQL database combined with a simple and well-documented API layer. This made it easier to integrate with the Express backend while still maintaining a structured relational database model.[52] Compared to alternative solutions such as Firebase or manually hosted SQL databases, Supabase offered a good balance between simplicity, flexibility, and control over relational data. Since our application relies heavily on structured relationships between users, clubs, and events a relational database system such as PostgreSQL was more appropriate than a purely NoSQL solution.[22]

Using supabase therefore allowed us to implement a database system that is both practical and aligned with modern web development. In addition to database functionality, Supabase is also used for storing image files related to

clubs via its built-in storage service. [53]

Authentication Approach for the system were instead of implementing a full authentication system with email and password login, a simplified authentication model. Each user is assigned a randomly generated user ID together with a role, which is stored in the database. Session handling is managed using express-session, allowing the application to maintain user state and determine access rights during navigation. [13]

This approach was chosen because authentication was not a primary goal of the system. The application is designed as an extension to the platform, Moodle, where users would already be authenticated before accessing the system. In such a context, login and identity management would be handled externally by Moodle, meaning that implementing a separate full authentication system within this application would be unnecessary. As a result, the focus was instead placed on core functionality such as data handling, role-based access, and interactions between users, clubs, and events.

3.2 Frontend

This section describes how the interface was structured and built using HTML, CSS, and JavaScript, and walks through the key architectural decisions that shaped the final result. Each page of the application is then presented in its own subsection.

Frontend Architecture Overview

The frontend of the application is built using a native web stack consisting of HTML, CSS and JavaScript, without the use of external frameworks. This approach was chosen to reduce complexity and maintain a clear understanding of the underlying browser technologies. To ensure that the project remains manageable, we implemented a modular folder structure based on the principle of Separation of Concerns, meaning that the structural, visual, and logical

parts of the application are kept in separate, dedicated directories.

All client-side files are located in the /public directory:

/components (HTML): This directory stores structural templates and reusable HTML fragments. Using template tags keeps the main pages clean and ensures that the structure is defined in HTML rather than hard coded into JavaScript strings. Reusable components are dynamically inserted into the DOM, reducing duplication and improving maintainability.

/CSS: This folder handles all styling, partitioned by components and user roles to prevent design conflicts. A custom CSS approach was chosen to maintain full control over the design and layout of the application.

/JS: This folder acts as the functional core, containing services for centralizing Fetch API calls for data consistency, and pages for handling logic for specific views.

/owner and /student: These directories serve as role based containers for HTML views.

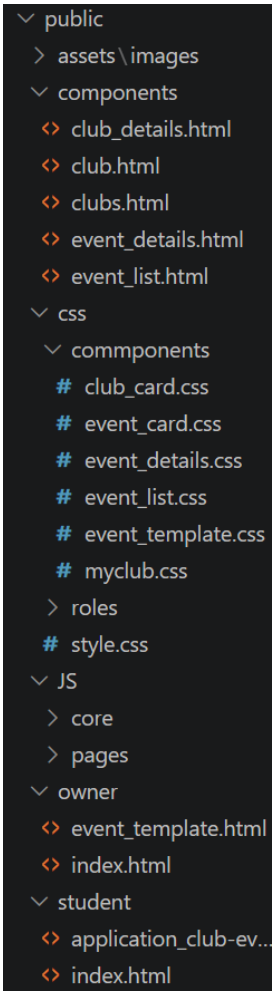


Figure 3.6: Frontend Folder Structure.

A key decision in the folder structure was to balance shared code with structural separation. For common elements like navigation bars and global styles, the application uses a shared page structure so we don't have to duplicate the same code across the site. In contrast, we chose entirely separate directories for the core dashboards. Combining these two different user views into a single file would make the code complex, requiring heavy JavaScript conditions just to show or hide elements based on permissions. Separate folders make the layout logic simpler and improve performance.

To secure these directories, the application utilizes role based routing. When a user attempts to access a specific dashboard, a background script automatically checks their saved user context before rendering the page. If an unauthorized

user tries to access a restricted folder, the script blocks the view and redirects them back to the login page.

Rendering and Data Fetch Flow

To create a fast, dynamic experience without a framework, the folders work together using an asynchronous data flow. The process follows the following order whenever a user opens a page:

1. Initialization: A script in /js/pages detects that the page has loaded and triggers a request

2. Asynchronous Fetch: The /js/services layer executes a fetch() request to the backend API and retrieves real time data as JSON. [49].

3. Template cloning: The page script finds the right HTML template inside the /components folder and clones it.

4. DOM injection: The script puts the data into the cloned template and inserts it directly into the page container using .appenChild(). This ensures smooth, fast and real time rendering without browser lag.

Color Recognition

Through CSS, each club has its own distinct color identity that is visible directly on the club list. These club colors are also used for color-coding events, making them easily recognizable on the calendar when a user joins an event. Furthermore, when a user creates a new event, they can choose a specific color for it. This use of color recognition gives users an instant, non-verbal way to distinguish between different clubs and schedules, which is intended to make the platform aesthetically pleasing, much easier to use, and to reduce navigation mistakes.

3.2.1 Login Page

This section describes the login page, which serves as the entry point to the application. Although the login does not implement full authentication, as this is assumed to be handled externally by Moodle. It establishes the user's identity and role within the system.

The login page serves as the initial entry point of the application. However, it does not function as a complete security mechanism, but instead as a simplified way of establishing user context within the system.

Design Rationale and Iteration

The visual design of the login page follows a clean and minimal structure inspired by Aalborg University CPH and Moodle design principles. This choice was made to ensure visual consistency with familiar institutional systems, which improves user recognition and reduces the learning curve. Recognition rather than recall[36] reduces cognitive load in user interfaces, while aesthetic and minimalistic design[36] improves clarity and focus.

The interface uses a centered login container, creating a clear visual hierarchy where the login box naturally becomes the focal point of the page and supports ease of navigation.[37]

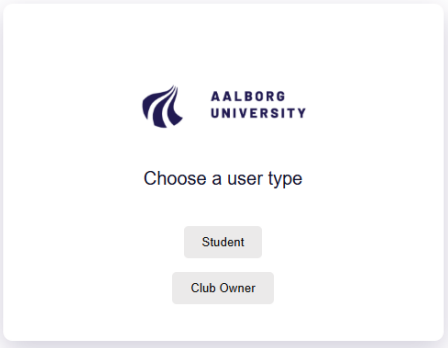


Figure 3.7: Implementation - login page

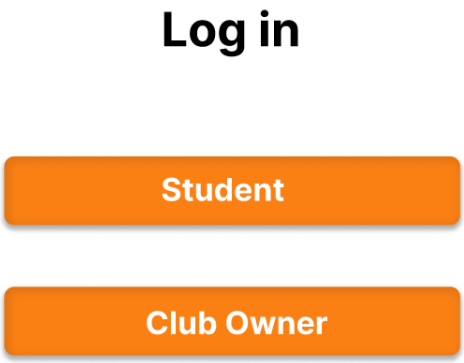


Figure 3.8: Hi-fi prototype - login page

The design was revised from an earlier prototype that used an orange-based color scheme (See figure3.8). The updated white and blue themed version was implemented to improve visual consistency with Aalborg University CPH branding and Moodle-style interfaces (See figure 3.7). This redesign strengthens system consistency and makes users more likely to understand and interact confidently with familiar interface patterns.

User Interaction and System Flow

When a user selects a role on the login page, the frontend sends a POST request to the backend using the Fetch API [31]. This initiates the login flow and establishes the user context within the application.

The frontend is responsible for handling user interaction and interface behavior, while the backend manages data processing, session handling, and persistent user storage. This separation of concerns improves maintainability and ensures a clear distinction between presentation logic and application logic.

The backend implementation of the login flow is further described in the backend architecture section 3.3.

3.2.2 Calendar

This section describes the calendar feature, which serves as the main landing view of the platform. The calendar provides students with a weekly overview of upcoming events and is central to addressing the problem of fragmented information identified in the problem analysis

The calendar directly addresses the issue of fragmented information identified in the problem analysis[6]. By gathering events in a single visual overview, the system reduces the need for students to search across multiple platforms This supports the overall goal of improving accessibility and making social activities more visible to students [20].



Figure 3.9: Example of calendar and modal window

Design Considerations

The visual design of the calendar in the Clubs Events extension is inspired by Moodle's CalMoodle, which is a weekly calendar displaying the current weekdays with corresponding dates, and upcoming lessons that are color-coded based on course. These lessons can be interacted with, causing a modal window to appear displaying the necessary course information.

In the Clubs Events calendar, the lessons have been replaced with color-coded events. Instead of course information, the modal windows display relevant event information together with the possibility of signing up for the selected event. Additionally, the weekdays have been expanded to include weekends, as events may take place outside regular weekdays.

This design choice, similar to previous design decisions, was made to keep the interface familiar to what the users are already accustomed to. It builds on the previously mentioned heuristics, particularly consistency with existing systems and recognition of familiar interface patterns[36] .

System structure

The calendar uses Monday as a fixed reference point, making it possible to calculate which events belong to the currently displayed week, as well as which day they should be placed on within the calendar's grid structure. This makes week navigation more stable and simplifies the placement of events in relation

to the weekdays.

When the user navigates back and forth between calendar weeks, the calendar's week offset changes, after which the calendar content is re-rendered based on the new reference week.

When a club-owner creates an event, the event information is entered through a front-end form containing details such as title, date, time, location and description. Submitting the form sends a POST-request to the backend API endpoint /events, where the request is processed through routes and controllers before the event is stored in the database.

When the calendar page is loaded, the front-end sends GET-requests to retrieve the stored events from the database[31]. The returned data is then used to dynamically generate the calendar content through JavaScript and DOM manipulation. HTML elements are created using functions such as createElement() and inserted into the page using appendChild(), making it possible to display events without predefined HTML content.

Before every re-rendering, the existing event-container is cleared to avoid events from previous weeks remaining visible. This also ensures synchronization between the application state and the user interface, preventing duplicate event elements from appearing during navigation and improving system reliability. The retrieved events are then filtered according to the currently displayed week and inserted into the calendar. Dynamic rendering was chosen to ensure that calender content automatically reflects database changes without requiring manually predefined HTML Structures[21].

Handling date placements required careful synchronization between JavaScript date calculation and CSS Grid positioning to ensure events appeared correctly across week transitions. The events' dates and times are used to calculate their placement within the calendar's CSS Grid layout. Times are converted into grid-rows, while weekdays determine the correct grid columns, allowing events to appear at their precise time and date within the interface.

Each event is assigned an event listener, making it possible for users to interact directly with events in the calendar. Interacting with an event opens a modal window displaying additional information together with the possibility of signing up for the selected event. Using modal windows allows users to access event details without leaving the calender overview, reducing navigation interruptions, and maintaining contextual awareness[37].

Figma

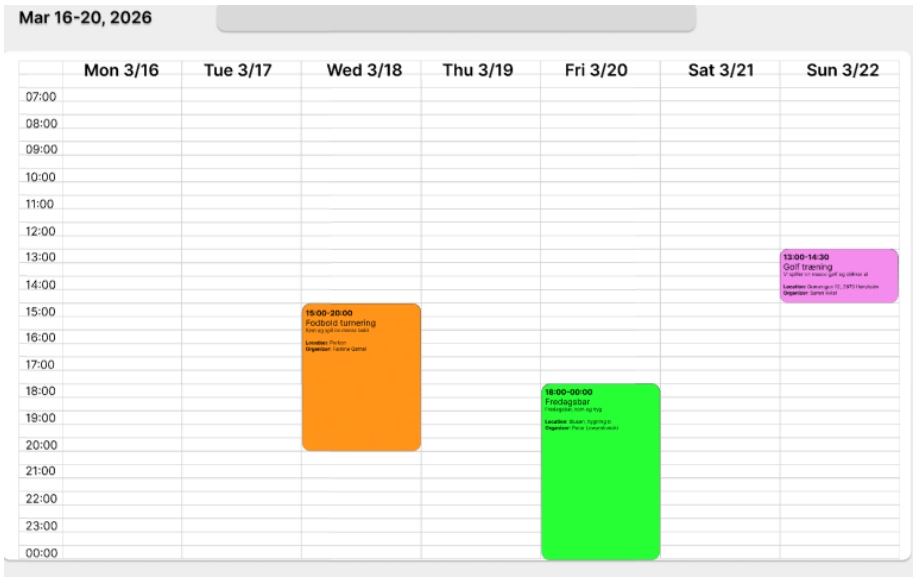


Figure 3.10: Figma calendar prototype.

The final calendar ended up pretty close to our Figma in terms of how it works, with the weekly grid, color-coded events, and modal windows all making it into the final version. The biggest thing you'd notice comparing the two is the colors. In Figma we went with really bright and vivid colors for the events, but when we built the actual thing we toned it down a lot to better match Moodle's look and feel. That was a deliberate choice since we wanted it to feel like it actually belonged in Moodle rather than looking like a completely different product. Otherwise most of the things are the same.

3.2.3 Club list

This section describes the club list page, which gives users an overview of all available student clubs on the platform. It covers how club data is fetched and rendered, how the page adapts to different user roles, and the design decisions made to ensure the list is easy to navigate and visually consistent with the rest of the system.

Design Considerations

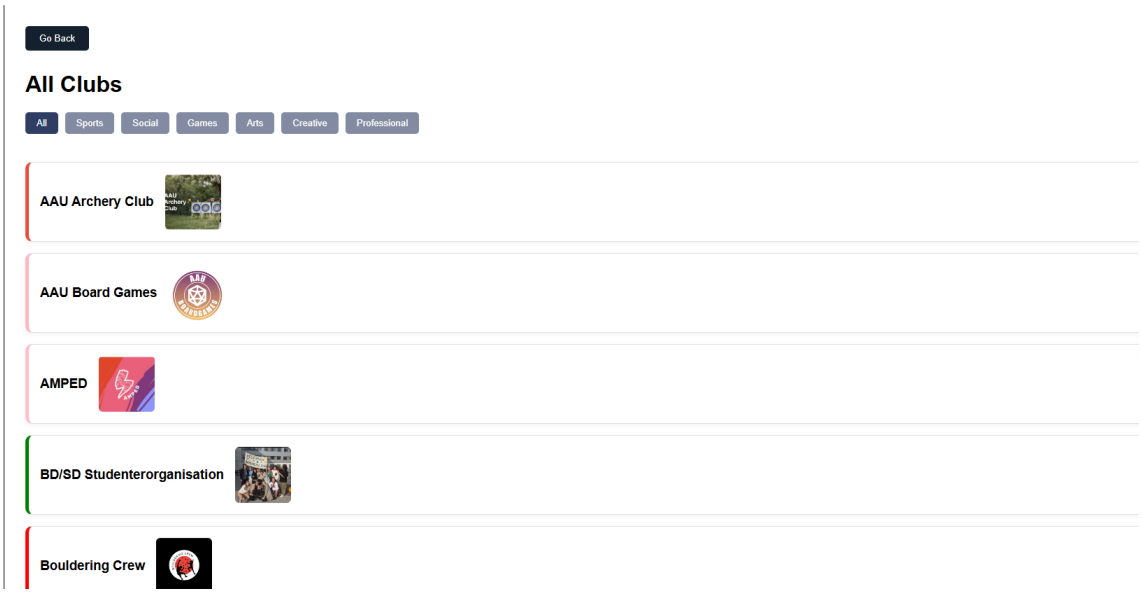


Figure 3.11: Club list page

Each club is displayed as a card constructed by cloning an HTML `<template>` element. This keeps the HTML structure centralized and avoids manual construction of DOM strings, improving maintainability and reducing duplication. Cards display the club name and an optional image. If no image is provided, the image element is hidden automatically. Each card contains a 5px left border in the club's assigned color, giving each club a recognizable visual identity. The same color is also used for club events to create consistency between clubs and their related activities. Club owners are visually identified through a blue "My Club" badge rendered next to the club name. In addition, all clubs contain categories that are inserted dynamically during rendering, allowing the

```
/*Get club info */
app.get("/clubs", async (req, res) => {
    const { data, error } = await supabase
        .from("clubs")
        .select("*")
        .order("name", { ascending: true });

    if (error) return res.status(500).json(error);

    res.json(data);
});
```

Figure 3.12: Code for fetching clubs

clubs to be filtered based on category. [32]

Club cards include hover effects in which the cards are slightly lifted upward, displaying a blue border and a light blue background tint[36]. This provides clear visual feedback that the cards are interactive without relying solely on cursor changes. The "Go to Top" button uses smooth scrolling to improve navigation throughout the page, so the user can orient and maintain their sense of location. [16]

System Structure

Clubs are fetched from the `/clubs` endpoint. The frontend calls the `getClubs()` function, which sends a request to `http://localhost:3000/clubs`. The backend route handler queries the database for all clubs and returns the data as JSON. The frontend then dynamically generates the club cards based on the returned data. The list refreshes automatically if the user navigates back to the page through the `pageshow` event.

The displayed content differs slightly depending on the user's role. Club owners are shown the "My Club" badge, while regular users are presented with the standard club view. The main difference occurs when selecting a club, where users are redirected either to the club owner's view or the student's view of the selected club. Both routes resolve dynamically to the correct dashboard URL based on the active user role. This structure allows the club list to function as a shared page rather than maintaining two separate implementations.

Requirements

The club list was categorized as a "Must Have" feature in the MoSCoW model. Its primary purpose is to provide users with an accessible overview of available clubs, simple navigation, and the ability to filter clubs by category. The color system was implemented to help users recognize connections between clubs and their events more easily. Although a search bar was not implemented in the final application, it was included as a "Could Have" feature for potential future development2.1.4.

Figma

The final implementation differs slightly from the original Figma prototype due to feedback gathered during usability testing. Features such as color-coding for distinguishing clubs, displaying full club names, and replacing the search bar with a filtering system were introduced based on user feedback. Furthermore, the clubs are sorted alphabetically instead of by creation time, making the list easier to navigate and search through for users.

3.2.4 Create Event

This section describes the event creation feature, which allows both students and club owners to add new events to the platform. Because the two user roles have different needs and levels of access, the feature is implemented as two separate forms.

Design Considerations

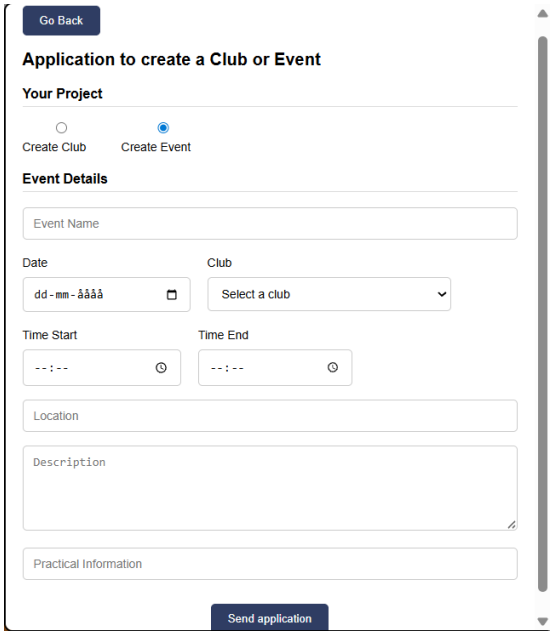


Figure 3.13: Create event form for students

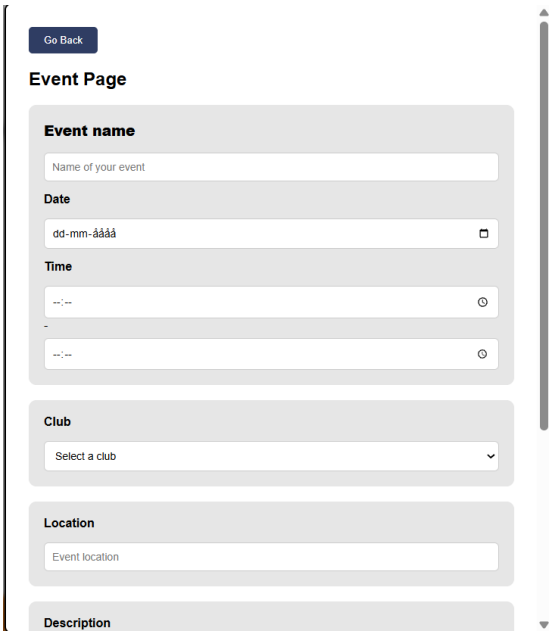
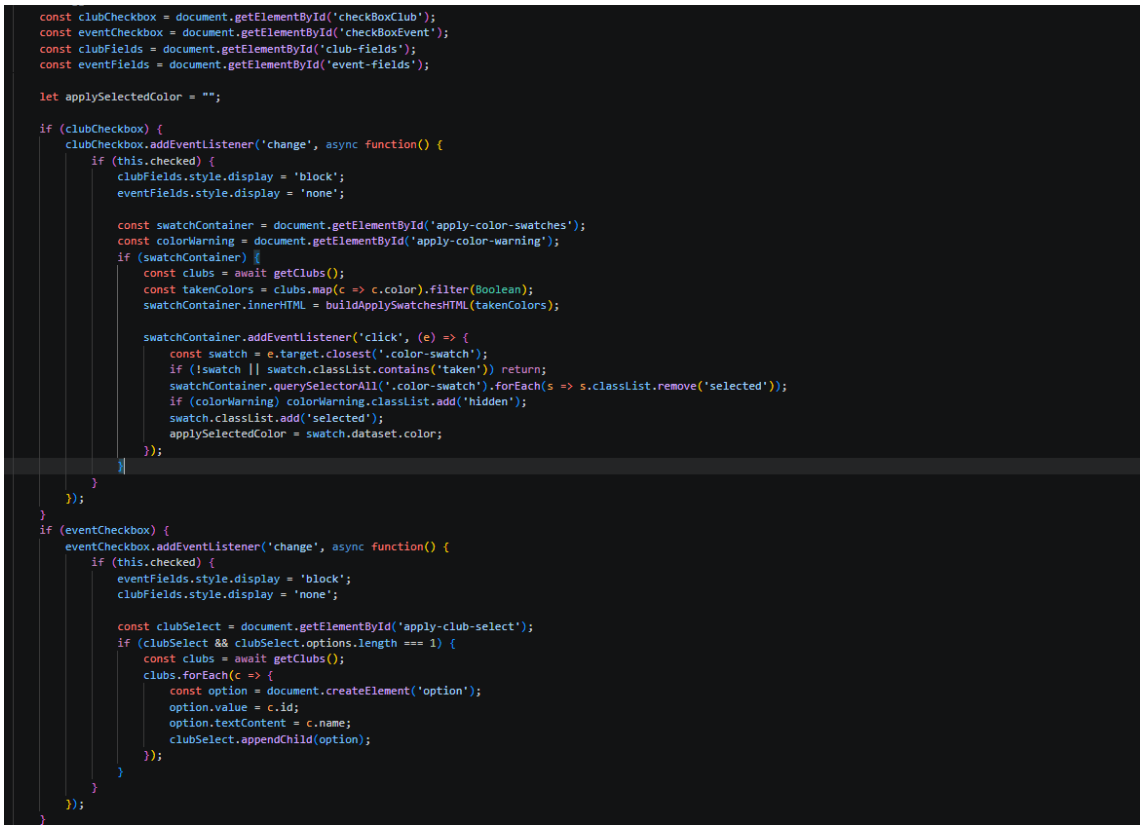


Figure 3.14: Create event form for club owners

Figures 3.13 and 3.14 illustrate the two separate entry points for event creation. The student modal and the club owner form are kept distinct to avoid unnecessary complexity for either role.

Instead of one shared form, the create event feature is implemented as two different forms, depending on the user's role. The "Create Event" button will take club owners to a dedicated full-page form. Students use a modal that allows for club and event creation in the same dialog. The user toggles between the two using radio buttons. This way, neither 'form' has the complexity of the other, and role-specific behavior does not have to be conditionally hidden or shown inside a single shared component. The student modal combines the two creation types into a single dialog, instead of two modals, so a student who changes their mind doesn't have to close one and open another. They appear slightly different because we wanted to match the student end of the creation of events with the creation of clubs in style. [9]

System Structure



```
const clubCheckbox = document.getElementById('checkBoxClub');
const eventCheckbox = document.getElementById('checkBoxEvent');
const clubFields = document.getElementById('club-fields');
const eventFields = document.getElementById('event-fields');

let applySelectedColor = "";

if (clubCheckbox) {
    clubCheckbox.addEventListener('change', async function() {
        if (this.checked) {
            clubFields.style.display = 'block';
            eventFields.style.display = 'none';

            const swatchContainer = document.getElementById('apply-color-swatches');
            const colorWarning = document.getElementById('apply-color-warning');
            if (swatchContainer) {
                const clubs = await getClubs();
                const takenColors = clubs.map(c => c.color).filter(Boolean);
                swatchContainer.innerHTML = buildApplySwatchesHTML(takenColors);

                swatchContainer.addEventListener('click', (e) => {
                    const swatch = e.target.closest('.color-swatch');
                    if (!swatch || swatch.classList.contains('taken')) return;
                    swatchContainer.querySelectorAll('.color-swatch').forEach(s => s.classList.remove('selected'));
                    if (colorWarning) colorWarning.classList.add('hidden');
                    swatch.classList.add('selected');
                    applySelectedColor = swatch.dataset.color;
                });
            }
        }
    });
}
if (eventCheckbox) {
    eventCheckbox.addEventListener('change', async function() {
        if (this.checked) {
            eventFields.style.display = 'block';
            clubFields.style.display = 'none';

            const clubSelect = document.getElementById('apply-club-select');
            if (clubSelect && clubSelect.options.length === 1) {
                const clubs = await getClubs();
                clubs.forEach(c => {
                    const option = document.createElement('option');
                    option.value = c.id;
                    option.textContent = c.name;
                    clubSelect.appendChild(option);
                });
            }
        }
    });
}
```

Figure 3.15: Code of student modal toggle

The forms collect the same core data: event name and date. Some of these features are a requirement to fill in, like name, date, start, and end time. We found out through our surveys that this information is important for student participation (See appendig A.1 and A.2).

Figure 3.15 shows the toggle logic used in the student modal, allowing the user to switch between club creation and event creation within the same dialog using radio buttons.

The club drop-down is dynamically populated by calling getClubs() when the event form is opened, rather than on initial page load. This keeps the modal light and ensures club data is only fetched when it is needed. We used a dropdown for club selection to ensure users can only select from valid existing clubs, rather than manually typing input. This reduces the risk of invalid or inconsistent data (e.g., spelling variations or non-existent clubs) and improves data integrity on the backend.[36]

The information inputs that are included in our event form are start time, end time, club (selected from a dropdown), location, description, and practical information. All fields are client-side validated before anything is sent. Whitespace is trimmed from values on submission, and the club ID is explicitly cast to a number from a string. Events are always created with isPublished: true. There is no draft state, which keeps the data model simpler and the user flow shorter. The payload is sent to POST /events.

Date and time functions use the browser's native HTML inputs, which handle formatting automatically. The form uses two distinct time inputs for start and end time, which is done to have both inputs function using the selection menu instead of just typing a time. This prevents the time from being over 24 hours; however, it doesn't apply to the minutes. It was also a choice to make it clearer for the user when their event should start and end by keeping them separate. After the backend will then merge the two into a single string for storage. If an event is later edited, the frontend splits that string back apart on the " - " separator to repopulate the two fields.

During submission, the submit button is disabled to prevent multiple requests to the backend. A status message element cycles through states: "Saving event..." while the request is in flight, then either "Event saved." or an error message from the backend. This message will be shown to the user to clearly show if an event has been published or not.[20]

On success, the form or modal is hidden, and the page reloads. Rather than manually inserting the new event into the DOM, the reload reuses the existing fetch-and-render logic so the new event appears in the list exactly as it would on a fresh load.

Requirements

Our core requirements were filled out for the event creation form, but the requirement for a club owner to have a repeated event, which is a "Could have", is not implemented which is mentioned in section 2.1.4. Having a button for choosing if an event should be recurring would improve usability for the club owner, because then they would not have to recreate the event each week.[6]

Figma

Our Figma is very similar to our event creation, with slight differences in having an end and a start time instead of just one. The biggest difference is that events are tied to clubs when creating events. This is done because we wanted to have our events appear under a club, so students could have easy accessibility to all the events by a specific club. Having an easy overview fits with our overall problem statement and was something we valued to have as a feature.[37]

3.2.5 Event List

This section describes the event list page, which provides users with a scrollable overview of all upcoming events on the platform. The event list was added as

a complement to the calendar view, giving users a more direct way to browse and filter events without navigating through specific dates.

The event list page gives users a straightforward way to browse upcoming events on the platform. It sits alongside the weekly calendar as a complementary view, useful for anyone who wants to scan ahead or find something specific without navigating through dates. The calendar works well for a time-based overview, but a scrollable list is simply easier when you are looking for a particular activity or planning further out

Design Considerations

The event list interface was designed with a strong focus on simplicity and efficient navigation. Similar to the other pages in the system, the interface follows a structured and minimalistic layout inspired by Moodle. This design approach was chosen to create a familiar environment for students and reduce unnecessary visual complexity, allowing users to focus more easily on event related information. This follows Nielsen's usability heuristics, especially Aesthetic and Minimalist Design and Consistency and Standards [37].

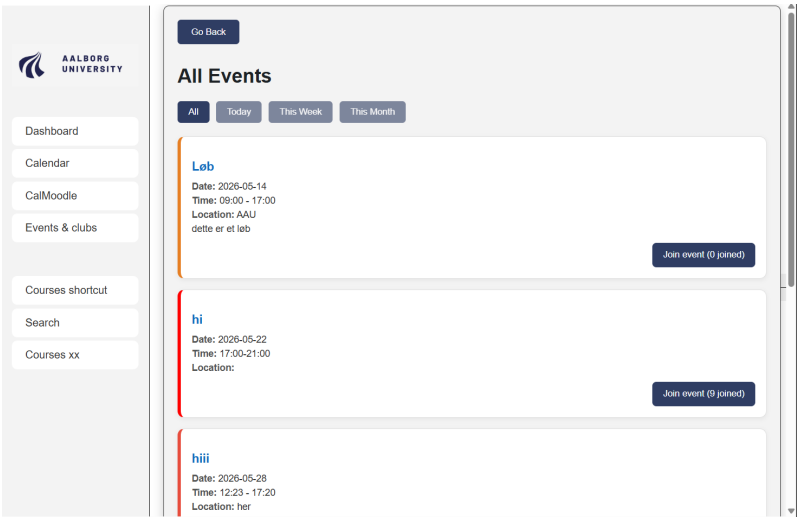


Figure 3.16: Implementation of the Event List

Consistent button placement and aligned event cards were used to create a more structured and intuitive interface. The layout uses clear alignment, spac-

ing, and proximity to group related information together and improve readability [6]. Interactive elements such as filter buttons and join buttons use hover effects and active states to give users visual feedback and create a more responsive user experience [20].

System Structure and Backend Logic

In the event list page, when the user opens the event list interface, the frontend requests event data from the backend using the Fetch API. This allows the frontend and backend to communicate dynamically, so the event list can update without requiring the user to reload the page. The request is handled through the following frontend function:

```
12  export async function getEvents() {
13      const res = await fetch("/events", { cache: "no-store" });
14      return await res.json();
15  }
```

Figure 3.17: Frontend request – Event list.

The request is handled by an Express.js GET route in the backend. GET routes are used to retrieve data from the server and send it back to the frontend [56]. In this implementation, the backend retrieves event information such as event title, date, time, location, description, and publication status from the Supabase database. In addition, related club color data is also retrieved and used for the visual styling of the event cards.

```
app.get("/events", async (req, res) => {
    const { data, error } = await supabase
        .from("events")
        .select(`
            *,
            clubs(color)
        `);

    if (error) return res.status(500).json(error);

    res.json(data);
});
```

Figure 3.18: Event retrieval from backend

```
const filteredAndSorted = events
  .filter(event => {
    const eventDateTime = new Date(`${event.date} ${event.time.split("-")[0]}`);
    return eventDateTime >= now;
  })
  .sort((a, b) => {
    const dateA = new Date(`${a.date} ${a.time.split("-")[0]}`);
    const dateB = new Date(`${b.date} ${b.time.split("-")[0]}`);
    return dateA - dateB;
  });
```

Figure 3.19: Frontend filtering and sorting – Event list

After the event data has been retrieved from the backend, the frontend processes the events before displaying them in the interface. The system filters out expired events and sorts the remaining events chronologically to create a more structured event overview. This functionality is implemented in the event list through the filtering and sorting logic shown above. The processed events are then dynamically rendered as event cards in the interface, where each card displays the previously described event information for the user.

Figma

The event list was actually not part of our original Figma prototype. During the high fidelity prototyping phase, events were only accessible through the calendar interface. However, the usability testing showed that several users preferred a more direct and scrollable overview when searching for events, as described in Section 2.3. This led us to add the event list as a separate page in the final implementation. It was a direct response to user feedback rather than something we had planned from the start.

3.2.6 Information Page

This section describes the club information page, which displays all relevant details about a specific student club. When a user selects a club from the club list, they are directed to this page. The page displays all the necessary details, including:

- Club information: The name of the club, a description of what they do, and a hero banner image.
- Time and location
- Contact information: E-mail and phone number.
- A counter showing how many students have joined the club.
- Upcoming events: A list of future events hosted by the specific clubs.

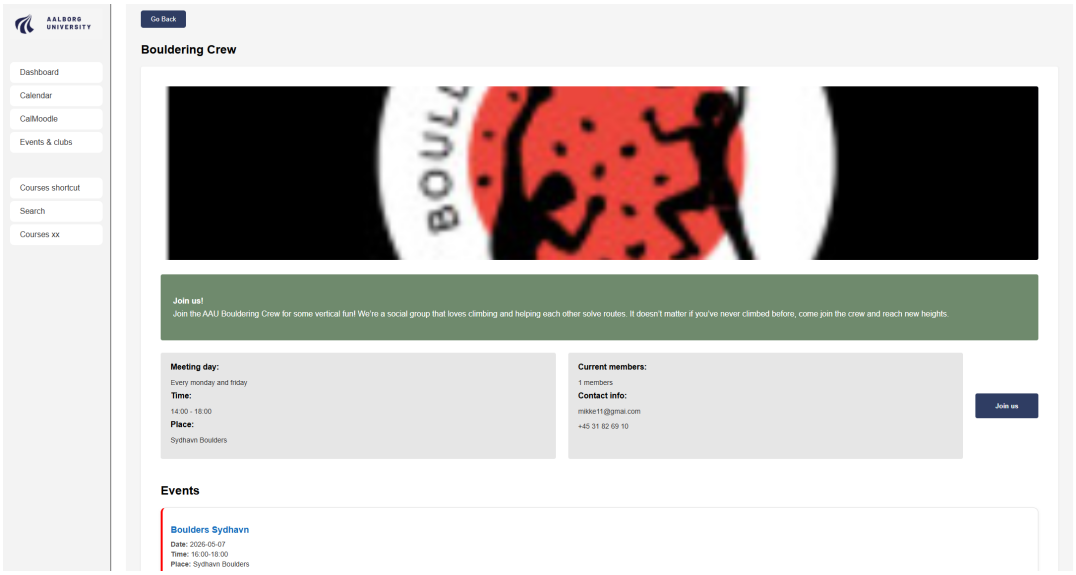


Figure 3.20: Information Page of a Club.

Figure 3.2.5, shown above, displays the information page of a club. Creating an information page was a core functional requirement of our project. One of the main issues identified at Aalborg University CPH was the lack of accessible information about the social events and clubs. Because of this problem, it was important to gather all relevant details in one place so students wouldn't have to struggle to find basic information.

Design Considerations

The page is styled to look clean, minimalistic and to match the design principles of Moodle. Since students use Moodle daily, making our application look similar makes it instantly familiar and easy to navigate. Physically, the structure uses a clear hierarchy inside a main outer box called ".white-box". At the top, the page features a large header image (.hero) to highlight the club's visual identity. Below the header, the club details are organized into two side-by-side columns using ".info-card" layouts. At the bottom of the page, the event section displays cards for upcoming events in a clean list row.

System Structure

The backend data architecture is supported by a frontend code structure split

into structural markup (HTML) and behavioral logic (JavaScript). Together, these files read the browser context, fetch data dynamically from Supabase via our backend routes, and build the information page layout. Instead of creating a separate HTML file for every single club, the website uses one reusable layout skeleton (club-details.html). When the page loads, JavaScript automatically pulls the correct data for that specific club and dynamically fills in the components.

When the pages opens, JavaScript grabs the club's unique ID from the browser's address bar. Instead of fetching the club details and member counts one after another, it uses "Promise.all" to fetch both from the database at the same exact time. This makes the page load much faster. The code for this is shown in Figure 3.21 below:

```
const params = new URLSearchParams(window.location.search);
const clubId = params.get("id");
const container = document.getElementById("club-page-content");
const userRole = await getUserRole();
const isOwner = userRole === "club_owner";
const dashboardUrl = isOwner ? "/owner/index" : "/student/index";

document.getElementById("eventsAndClubsLink")?.addEventListener("click", () => {
    window.location.href = dashboardUrl;
});

if (!clubId) {
    container.innerHTML = "<p>No club specified.</p>";
    return;
}

const [clubs, events, members] = await Promise.all([
    getClubs(),
    getEvents(),
    getJoinCount(clubId)
]);
```

Figure 3.21: Sudent and Club Owner roles.

The system automatically checks if the logged-in user is a student or the club owner. This dictates what buttons they can view and interact with. For

students, the script unhides the main “Join us” button (join-btn), It then runs an external component function (setupClubJoinButton) that attaches the database registration logic, allowing students to instantly join the club. If the user is a club owner, the join button stays hidden. Instead, the logic displays the “Edit Club” button and automatically pre fill the administrative editing form fields with the club’s current data.

To keep the main markup clean and organized, the project utilizes structural HTML components via template tags:

```
<template id="event-card-template">
    <div class="event-card">
        <div class="event-info">
            <h3 class="event-title"></h3>
            <p><strong>Date:</strong> <span class="event-date"></span></p>
            <p><strong>Time:</strong> <span class="event-time"></span></p>
            <p><strong>Location:</strong> <span class="event-location"></span></p>
            <p class="event-description"></p>
        </div>
        <div class="event-actions">
            <button id="join-event-eventlist" class="button  hidden join-event-button">Join event</button>
        </div>
    </div>
</template>
```

Figure 3.22: Template Tags.

Using these HTML templates as shown in Figure 3.22, enforces a clean Separation of Concerns. Instead of cluttering the JavaScript code with long, hard-coded layout blocks, these predefined HTML blueprints reside strictly within the HTML file, The JavaScript code simply fragments or clones them when needed, filling them safely with text content before injecting them into the DOM.

Requirements

The information page was one of our Must have features since one of the core problems we identified at AAU was that students couldn’t easily find detailed information about clubs and events. Having a dedicated page that gathers everything in one place, like contact info, upcoming events, and a description of the club, directly addresses that problem. Without it, the platform would just be a list of club names with nowhere to go, which wouldn’t solve much.

Figma

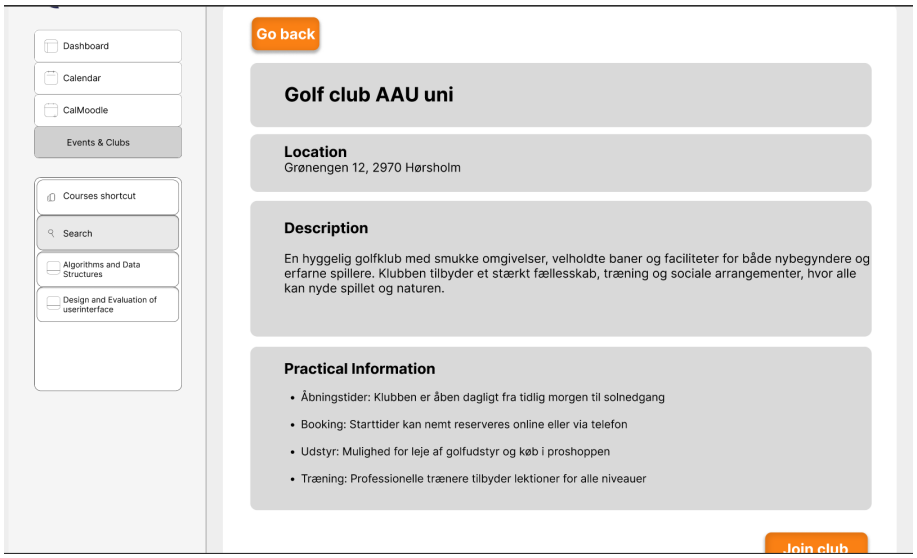


Figure 3.23: Information page prototype of a club.

Figure 3.23, shown above, displays the club information page from our Figma prototype. The information page ended up pretty close to what we had in our Figma in terms of the content shown, things like club name, description, location, contact info, and upcoming events. The biggest change was the layout and colors. We went from a two column layout to a card based design with neutral gray tones, mostly because we wanted it to feel like it actually belonged in Moodle rather than looking like a completely different product. We also added the option for clubs to upload their own picture, which we felt was important to give each club its own identity and make the page feel a bit more personal. We also added a section showing the upcoming events for that specific club directly on the page, which makes it a lot more convenient for the user since they no longer have to navigate back and forth between pages just to see what a club has coming up.

3.3 Backend

This section describes the backend of the application, which is responsible for handling requests, managing sessions, communicating with the database, and serving data to the frontend. The backend is built using Node.js and Express.js

Backend Architecture Overview

The backend of the application is built using Node.js and Express.js, which serves as the main server framework.[39, 12] It is responsible for handling HTTP requests, managing sessions, communication with the database, and serving frontend pages.

The backend follows a modular architecture based on separation of concerns, which improves maintainability, scalability, and code readability.[21] The system is structured into dedicated folders for routing, application logic(controllers), middleware, and database configuration (See figure 3.24).

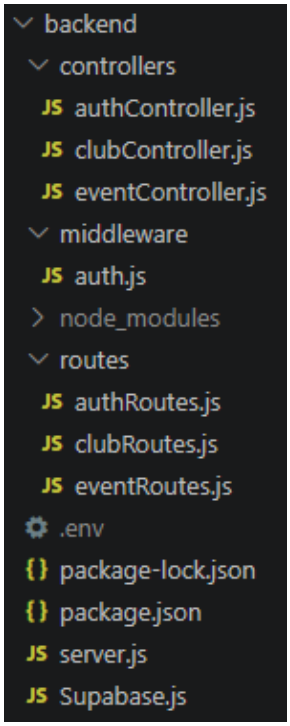


Figure 3.24: Backend Folder Structure.

The main entry point of the backend is the server.js file. This file is responsible for initializing the Express server application, configuring global middleware, registering route modules, and starting the server. It does not contain core application logic, which is instead delegated to separate modules. Compared to a monolithic backend architecture, modular architecture improves maintainability by separating responsibilities into independent components. This structure improves scalability, as each component can be developed and maintained independently without affecting the overall system.

Projet Setup and Dependencies

The backend project is managed using Node.js and the npm package manager. [41] All dependencies required for the application are defined in the package.json file.

Key dependencies include:

- Express.js - backend framework for routing and server logic
- Supabase JS client - communication with the database
- express-session - session management
- multer - file upload handling
- dotenv - environment variable management

When the project is installed, npm automatically generates the `node_modules` folder, which contains all external libraries required by the application. This folder is not manually maintained but generated based on package.json.

Sensitive configuration values such as API keys are stored in a .env file, which is loaded using the dotenv package.[40] Separating sensitive configuration data from the source code follows common security practices for secret management. [42]

Supabase Database Design

The application uses Supabase as a cloud-based PostgreSQL database.[52, 54] The database is designed using a relational model, consisting of multiple interconnected tables representing users, clubs, and events (See figure 3.25). [44]

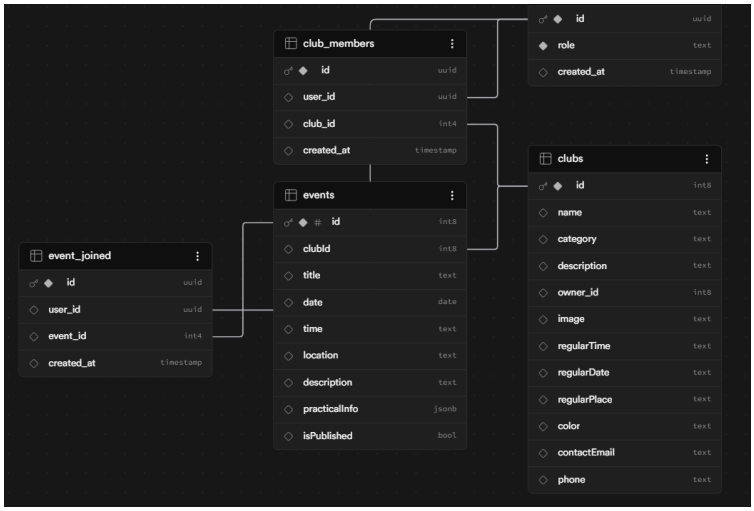


Figure 3.25: Supabase database schema.

The main tables are:

- users- stores user identity and role
- clubs - stores club information and metadata
- events - stores event data linked to clubs if related
- `club_members` - junction table for user-club relationships
- `event_joined` - junction table for user-event relationships

The relational design allows efficient management of many-to-many relationships. For example, a user can join multiple clubs, and each club can have multiple members. This is handled through the `club_members` table instead of storing arrays or duplicated data inside a single table. Foreign key constraints ensure referential integrity and automatically maintain consistency when records are deleted. [45]

This relational approach was chosen because the application is highly dependent on structured relationships between entities. A relational database such

as PostgreSQL is therefore more suitable than a NoSQL solution, which is less effective for relational data. [22]

Supabase Integration

Supabase is integrated through a dedicated configuration file (Supabase.js), which creates and exports a single database client instance used across the entire backend (See figure 3.26).

```
import 'dotenv/config';

import { createClient } from "@supabase/supabase-js";

const supabaseUrl = "https://sjtapuesjqubmdawxwzm.supabase.co";
const supabaseKey = process.env.SUPABASE_SERVICE_KEY;

export const supabase = createClient(supabaseUrl, supabaseKey);
```

Figure 3.26: Supabase.js file.

The design ensures that the database connection is centralized and reused across all modules, improving consistency and maintainability. It also avoids redundant initialization of multiple database clients.

Supabase was chosen over alternatives such as Firebase or self-hosted SQL database due to its combination of ease of use, scalability, and built-in PostgreSQL support. Additionally, Supabase provides integrated storage functionality, which is used for handling image uploads within the application. [53]

Routing Architecture

The backend API is structured using Express routing [12]. Although the system was not explicitly designed as a REST API, it follows REST-inspired principles through the use of standard HTTP methods[34]:

- GET - retrieving data
- POST - creating data

- PATCH- updating existing data
- DELETE - removing relationships

This approach ensures consistency and predictability in how data is handled across the system. Routes are separated into individual files based on functionality: authentication, clubs, events. This improves readability and ensures that each module is responsible for a specific domain of the application. In the following figure 3.27 the clubRoutes.js file can be seen, here all routes related to clubs are located. Such as fetching the joined count, creation of a club, or updating a club.

```
import express from "express";
import multer from "multer";

import { getClubs, getClubJoinCount, joinClub, leaveClub,
         createClub, updateClub, uploadClubImage } from "../controllers/clubController.js"

const router = express.Router();
const upload = multer({ storage: multer.memoryStorage() })

router.get("/", getClubs);
router.get("/:id/join-count", getClubJoinCount);
router.post("/:id/joined", joinClub);
router.delete("/:id/joined", leaveClub);
router.post("/", createClub);
router.patch("/:id", updateClub);
router.post("/:id/image", upload.single("image"), uploadClubImage);

export default router;
```

Figure 3.27: Example route file.

Controllers and Application Logic

Controller files contain the core application logic of the backend. They handle incoming requests, interact with the database, and return responses to the frontend. Separation of controllers from routes ensures a clear distinction between request handling and application logic [21]. Routes define which endpoint is accessed, while controllers define what functionality is executed when the endpoint is triggered. This separation improves maintainability, scalability, and makes the system easier to extend and debug.

The authentication system is implemented in a simple mechanism through a ”loginDemo” controller function (See figure 3.28). Instead of a full credential-

based authentication system, the application assigns each user a randomly generated UUID using Node.js' built in crypto module and a selected role [38].

When a role is selected on the frontend login page, a POST request is sent to the backend login endpoint using Fetch API[31]. The controller then initializes the user session, generates a unique UUID, and stores the user identity and selected role in the database. This flow illustrates the separation of conserns between the frontend and backend responsibilities.

```
export const loginDemo = async(req, res) => {
    const userId = randomUUID();

       req.session.user = {
        id: userId,
        role: req.body.role
    };

    //save user in database
    await supabase.from("users").insert([
        {
            id: userId,
            role: req.body.role
        }
    ]);

    res.sendStatus(200);
};
```

Figure 3.28: LoginDemo controller.

This approach, as mentioned earlier, was chosen because the system is designed as an extension of Moodle, where authentication is assumed to be handled externally. Therefore, the login process only establishes user identity within the application rather than performing full authentication.

Controllers also manage the main features, including: creating and updating clubs, creating and managing events, handling join and leave functionality for clubs and events, and processing image uploads via Supabase Storage. These functions form the application logic and ensure that database operations are performed in a structured and consistent way.

Middleware and role-based acces control

Middleware functions act as an intermediate layer between incoming requests and backend logic [14]. They are used to validate conditions before allowing

access to protected routes.

The application implements role-based access control using middleware defined in auth.js (See figure 3.29).

```
export function requireRole(role) {
  return (req, res, next) => {
    if (!req.session.user) { //user is not logged in
      return res.status(401).send("Not logged in");
    }

    if (req.session.user.role !== role) { //loged in but has wrong role
      return res.status(403).send("Not allowed");
    }

    next(); // allowes as has correct role
  };
}
```

Figure 3.29: RequireRole middleware.

The middleware ensures that only users with the correct role can access specific routes. For example, student routes are restricted to users with the student role. Middleware improves code reusability by centralizing access control logic instead of repeating it across multiple routes.

Session Handling

Sessions are managed using express-session, which stores user session data on the server [13]. This allows the application to maintain user state across multiple requests without requiring the user role to be selected repeatedly. The session stores: user ID, user role. This information is used by middleware to control access to protected resources and determine user permissions throughout the application.

This session-based approach was chosen over token-based authentication because it is simpler to implement and sufficient for the project's scope, especially given the assumption that authentication is handled externally by Moodle.

Summary

The backend of the application is built around a modular Express.js architecture integrated with a Supabase PostgreSQL database. The system follows a clear separation of concerns between routing, controllers, middleware, and database configuration, which improves maintainability and scalability.

Key design decisions such as the use of session-based authentication, role-based access control, and a relational database structure were made to align with the assumption that authentication is handled externally through Moodle. This allowed the focus to remain on core application functionality, including club and event management, user interaction, and data relationships.

3.4 Unit Testing

To verify that the core backend functionality works as intended, a set of unit tests was written targeting the most critical database interactions in the system. This section describes the purpose and scope of the tests, presents the two main test cases implemented, and discusses what the results indicate about the reliability of the backend.

Unit testing is used to check that important backend functionality in the system works correctly [5]. In this project, the testing mainly focuses on database interactions related to events and clubs, including retrieving data and updating existing information. Testing the backend helps detect errors early and ensures that the system returns the expected data when communicating with the database. Unit testing also improves system reliability and makes the application easier to maintain [5].

Retrieving Data Test

The first test checks whether the backend correctly retrieves event data from the database. The test sends a GET request to the /events route and checks if the returned response gets the correct event information.

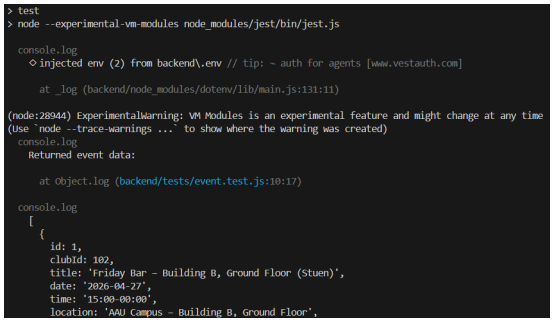


Figure 3.30: Backend event retrieval

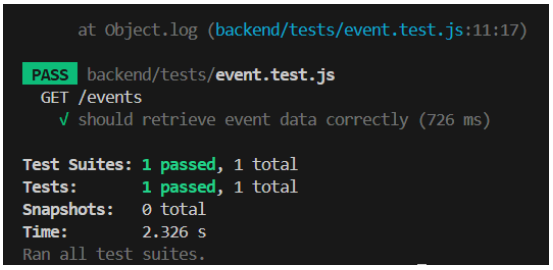


Figure 3.31: Backend test output

The test successfully retrieved all event data from the backend. It only shows a small example of the returned response, while the complete test output included all stored event information from the database. The returned data contained information such as event title, date, time, location, and more. The backend also returned the status code 200, confirming that the request was handled correctly and that the connection between the Express.js backend and the Supabase database works as expected.

Updating Data Test

The second test focuses on updating existing event data in the database. A PATCH request is sent to the /events/:id route with new event information, after which the test checks whether the changes were correctly saved and returned by the backend [46].

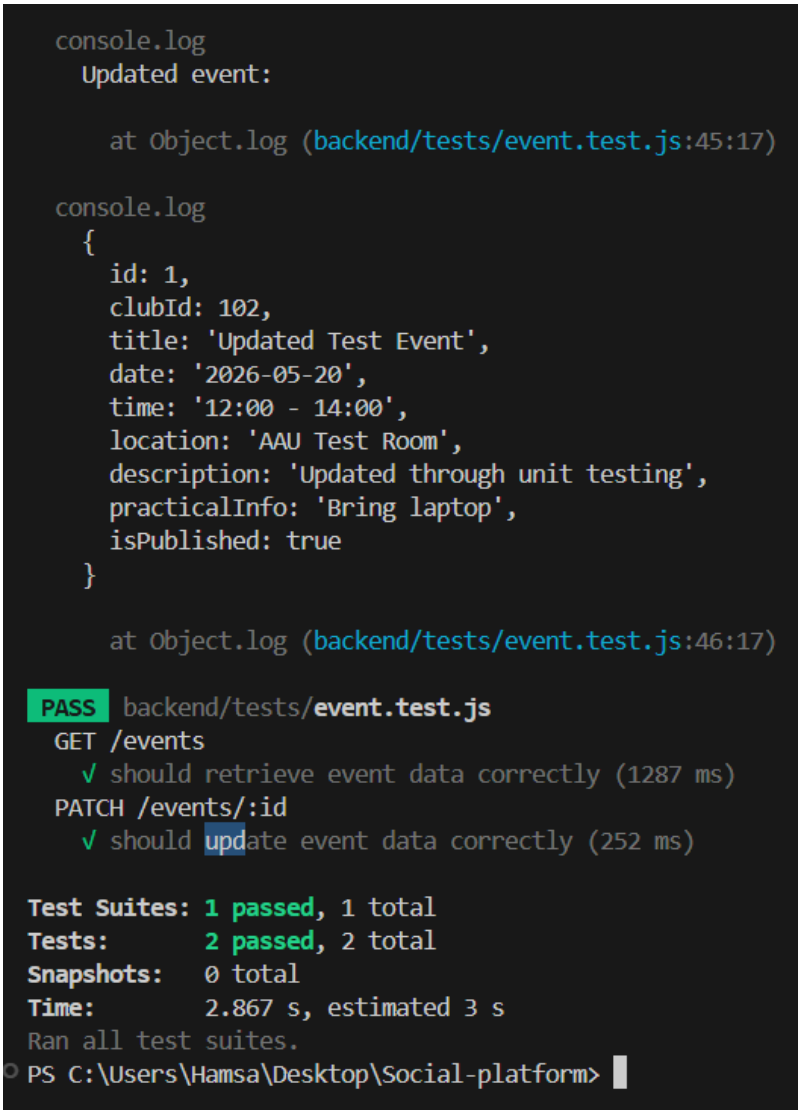


Figure 3.32: Updated Data from the backend

The test successfully updated the selected event in the Supabase database. The returned response showed the updated event information, confirming that the backend handled the PATCH request correctly and saved the new values in the database. The full unit testing code can be found in Appendix A.4.

Chapter 4

Usability Test of the Final Program

Following the implementation of improvements identified during the first usability test, a second test was conducted on the finished program. This section describes the setup and method used for the test, which followed the same Think Aloud approach and task structure as the first evaluation.

After implementing improvements based on the findings from our first usability test, a second usability test was conducted on the final impended program. The purpose of this test was to evaluate whether the updated design improved usability and to identify any other usability issues. The same tasks from the first usability test were reused to compare user interactions and evaluate whether the changes had a positive effect on the user experience:

- Log in to the platform as a student.
- Find a club you want to join and view its details.
- Create a new club or event.
- Find the number of attendees for the next event you are attending.
- Find an event you would like to attend and register your attendance.

The second usability test was also conducted using the Think Aloud method [35], where participants continuously verbalized their thoughts while interacting with the program. The test included three second semester students from the Software program at Aalborg University CPH, who represented the target group of the platform. During the sessions, one group member facilitated the test while another observed and documented user behavior and comments.

4.0.1 Analysis of Test Results and Feedback

The findings from the usability evaluation, along with the categorized severity levels of the participant feedback, are structured in Figure 4.1 below:

Task / Feature	Participant Feedback / Observed Behavior	Severity level	Recommendation for Future Iteration
Login System	Log in was described as straightforward and intuitive by all participants.	Positive	Maintain current design.
Navigation and Core flow	Users successfully navigated the calendar, joined clubs, viewed events, and created new events.	Positive	Maintain current layout and workflow.
Events Registration (Attendee Count)	The number of registered members only updated after reloading the page, whereas real time feedback was expected.	Serious	Implement automatic dynamic UI updates when a user registers for an event.
Club Membership ("Undo" Action)	The "Undo" functionality after joining a club was unclear and too closely integrated with the "Join" button.	Serious	Separate the "Undo" action into its own distinct, clearly labeled button.
Button Color Coding	The usage of colors was confusing: buttons with completely different functions ("Go Back" vs. "Send Application") looked too similar.	Serious	Apply highly distinct, contrasting colors to separate navigation buttons from primary action buttons.
List Visibility	The "Show Full Event List" and "Show Full Club List" buttons were visually overlooked by one participant because they appeared too small and unimportant.	Serious	Increase the size and visual weight of the list expansion button to improve discovery.
Calendar Event Pop-up	A participant noted that adding an edit function directly inside the calendar pop-up would improve flexibility.	Cosmetic / Minor	Add and "Edit" shortcut directly inside the calendar pop-up window.

Figure 4.1: Overview of usability test findings, categorized by feedback severity and recommendations.

As shown in the table above, the usability test on our final program demonstrated that the core flows, such as logging in, navigating the calendar, and creating events, are intuitive and well received. Overall, the second evalua-

tion also demonstrated that the design iterations implemented after the first evaluation significantly enhanced the overall user experience. However, the evaluation highlighted a few remaining usability challenges primarily related to feedback visibility and button design. These insights provide prioritized recommendations for future iterations of the platform.

Chapter 5

Discussion

5.1 Reflection on the Developed Solution

This chapter reflects on the project as a whole, examining to what extent the developed platform addresses the problem identified in the problem analysis. It discusses the key design decisions made throughout the project, evaluates the results of the usability tests, and considers the limitations of both the prototype and the development process.

The purpose of this project was to investigate how the communication of social activities at Aalborg University CPH affects student participation and to develop a digital solution that could improve accessibility to information about clubs and events. Based on the conducted research, questionnaire, interviews, usability evaluations, and implementation process, the project suggests that the developed platform addresses several of the key problems identified in the problem analysis presented in Section 1.1. Appendix A.2 Appendix A.1

One of the main issues identified throughout the report was that information about social activities is currently spread across multiple platforms such as Moodle, Facebook, Instagram, Discord, and Messenger. Both existing research and our own interviews showed that this fragmentation creates confusion and makes it difficult for students to maintain an overview of available activi-

ties. The implemented solution attempts to address this problem by gathering events, clubs, and related information into one centralized platform. Features such as the event calendar, event list, club overview, and standardized event pages directly support this goal by making information easier to locate and more structured.

The usability tests further indicate that the platform is generally successful in supporting the users' primary tasks. Participants were able to navigate the system, locate clubs and events, and sign up for activities without major difficulties. As discussed in Section 2.3, this suggests that the prototype may improve accessibility and visibility compared to the current communication structure. Additionally, design choices based on Nielsen's heuristics, such as consistency and recognition rather than recall, helped create a more intuitive experience by building on familiar Moodle-inspired interface patterns.

An important finding from the interviews was that students preferred a solution integrated into a platform they already use regularly. Several participants expressed that they often overlook information when it is communicated through separate social media platforms or external websites. This became an important non-functional requirement for the system: the platform should function as an extension of Moodle rather than as a completely separate website. Building upon Moodle may improve familiarity, reduce the learning curve, and increase the likelihood that students will use the platform because it becomes part of their existing university workflow. This requirement influenced both the visual design and the interaction flow of the final prototype.

However, the project also has several limitations. First, the system is developed as a proof of concept rather than a fully integrated production-ready solution. Although the prototype illustrates the core functionality and interaction flow, it does not currently operate directly inside Moodle. Instead, it simulates how such an integration could function. Features such as full authentication, messaging, and direct synchronization with Moodle systems are therefore simplified or excluded. In a real implementation, these systems would

need to be integrated with Aalborg University CPH's existing infrastructure.

Another limitation concerns the overall scope of the project. The initial intention was to follow a vertical prototyping approach introduced in Section 2.2, where the most essential features would be developed in depth before expanding the system further. However, during the development process, the solution gradually became more horizontal, as several different features were implemented to present the broader concept of a centralized platform. However, this also meant that some features, such as the join-button and others, were not developed to the same level of completeness and refinement.

Additionally, some navigation shortcuts remain unimplemented. For example, it would be convenient for the user to navigate directly from an event in the event list to the club which the event belongs. Similarly, users should ideally be able to access the corresponding club directly from an event displayed in the calendar view. These shortcuts could improve usability and create a more seamless user experience, but they were not completed within the scope of the project.

As a result, certain parts of the system should be understood as functional demonstrations rather than fully mature features. Some interactions support the intended user flow, but could benefit from clearer feedback, stronger validation, and more consistent behaviour across the platform. For example, the usability tests revealed a need for clearer feedback mechanisms, a better overview of memberships, improved discoverability of features, and more detailed event and club information. These features were categorized as Should have or Could have requirements defined in Section 2.1.4 and represent important directions for future development.

In hindsight, a narrower scope may have made it possible to develop fewer features more thoroughly. While the broad scope helped present the overall concept, it also made it more difficult to maintain a clear overview during implementation. This affected both the completeness of certain features and the consistency of the codebase.

Another limitation is that the usability tests were conducted with a relatively small test group consisting primarily of software students from Aalborg University CPH. This means the results may not fully represent the experiences and needs of the broader student population across different study programs. A larger and more diverse user group would likely provide additional insights into usability challenges and user expectations.

Technical Limitations

While the system successfully implemented the core functionality of a full-stack web application, several technical limitations were identified during development. These relate mainly to frontend scalability, backend architecture, database design, and long-term maintainability.

Frontend Limitations: The frontend was primarily designed for desktop use, meaning responsiveness across mobile devices is limited. Since many students frequently access university systems through phones or tablets, improving responsiveness would be important in future iterations.

Although the frontend was structured using separate folders and reusable components to maintain a clearer overview, some repeated styling and interaction patterns still appear across the application. This is especially visible in certain button designs, page-specific interactions, and smaller UI elements that could have been abstracted more consistently. As the prototype expanded, some design and code patterns were repeated instead of being centralized. A more consistent component-based approach could improve maintainability and make it easier to update the visual design across the platform.

Backend Limitations: The backend uses Node.js and Express.js in a modular structure presented in Section 3.1. However, some controllers still handle multiple responsibilities such as validation, database queries, and response formatting. This increases coupling and reduces scalability. A more robust design would introduce a service layer to separate application logic from routing and database operations, which could improve maintainability and extensibility.

Several backend features, particularly join and leave functionality for clubs and events, contain repeated logic with only minor differences. Both features follow similar patterns, differing mainly in table and field names. This indicates a lack of abstraction in the current backend design. While this approach simplified initial development, it also reflects how the overview of the codebase became more difficult to maintain as the scope of the prototype expanded. As more features were added, similar logic was implemented in multiple places instead of being extracted into shared functions. This reduces maintainability because changes to core logic must be applied in multiple locations. Extracting shared logic into reusable service functions could improve consistency and reduce duplication.

The relational database using Supabase is well suited for handling structured relationships between users, clubs, and events. However, one limitation exists in the manual generation of club IDs:

```
const nextId = maxData ? maxData.id + 1 : 1;
```

Figure 5.1: Manual club ID generation.

Although this approach functions correctly in a single-user or low-traffic environment, it introduces potential concurrency risks. If multiple requests are processed simultaneously, duplicate IDs could theoretically be generated. This reflects a trade-off between simplicity in application logic and robustness in database design. A more reliable approach would be to delegate ID generation entirely to PostgreSQL using identity columns.

Furthermore, the current implementation uses a simplified session-based role model because the project assumes that authentication would already be handled externally by Moodle. This is sufficient for the prototype, but a real implementation would require closer integration with Moodle's authentication and user management systems.

Overall, the project suggests that a centralized and Moodle-based platform has the potential to improve accessibility to information about social activities

and thereby support student participation and engagement. While the current system remains a prototype with several limitations, it illustrates how the identified communication challenges presented in Section 1.8 may be addressed through a more integrated and user-centered solution.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

6.2 Conclusion

The goal of this project was to investigate how the lack of accessible and centralized information about social activities at Aalborg University CPH affects student participation and to develop a digital platform that could improve accessibility and encourage engagement.

Based on the conducted research, interviews, surveys, usability evaluations, and implementation, the project addressed the identified problem statement. The findings throughout the report consistently showed that students experience frustration with fragmented communication spread across multiple platforms. This fragmentation makes it difficult to maintain an overview of clubs, committees, and events, which can reduce participation in the social environment at Aalborg University CPH.

To address this issue, a centralized prototype platform was developed as a Moodle-inspired extension that gathers social activities, clubs, and events into one place. The implemented features included an event calendar, event list,

club overview, information pages, and an event creation system for organizers. These features made it possible to provide a more structured overview of activities and reduce the need for students to navigate across several different platforms. In addition, the usability tests showed that users were generally able to complete central tasks such as locating clubs, finding events, and signing up for activities. This indicates that the platform supports the primary user needs identified during the problem analysis.

The project therefore reached several of its central goals. A functional proof-of-concept prototype was developed, the core Must have requirements from the MoSCoW analysis were implemented, and the solution was evaluated through usability testing with representatives from the target group. Furthermore, the project demonstrated how a more centralized communication structure may improve accessibility to information about social activities within a familiar Moodle-inspired environment.

However, some intended functionalities were not fully achieved. The system was not integrated directly into Moodle, several navigation shortcuts and interaction details remained incomplete, and the platform was only partially responsive on mobile devices. In addition, some Should have and Could have features were not implemented within the scope of the project. The usability testing was also conducted with a relatively small group consisting primarily of software students from Aalborg University CPH.

Overall, the project suggests that improving the visibility and accessibility of information about social activities through a centralized platform may contribute to a more connected and engaging university environment. At the same time, the project highlights several areas for future development, including full Moodle integration, improved responsiveness, broader usability testing, and further refinement of both frontend and backend functionality.

6.2.1 Future work

The main thing our program is missing, if it were to be constructed to be used in the real world, would be our implementation into Moodle. It would solve some of the features we are missing already to have in our program, like an individual user ID to tie the existing club to a club owner, rather than everyone being able to edit in all clubs as long as they are logged in as a club owner.

We have also discussed admin features. The role of an admin would be to give a student ownership over a club. Our students would make applications to create a club, and an admin would have the ability to deny a club or create it. This would be done to ensure that we don't have multiple of the same clubs(same thing would be true for events).

Security features would be something we would have to look into if we were to continue with the development.

Lastly, we would look at features that we didn't make in our "Could have" section, things like a search bar in our club list, a notification reminder that would tell you if your event was upcoming that you have signed up for, and a repeat weekly

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Appendix A

Appendix

A.1 Surveys

A.1.1 Question 1: Where do you usually look for information about social events at AAU?

- **Person 1** Primarily uses Facebook and Instagram (AAU Copenhagen). Finds it difficult to search for information elsewhere and prefers using one platform to make events easier to find.
- **Person 2** Usually sees what the tutor coordinator posts on Facebook. Only discovered Moodle Students during the second semester. Thinks it would be beneficial if information was not spread across multiple platforms and suggested displaying events on campus screens.
- **Person 3** Does not actively search for events because information is not gathered in one place, but would probably use Facebook if needed. Knows about Moodle Students, but finds it demanding since students must enroll themselves. Would like events to be communicated more clearly.
- **Person 4** Receives information directly from others and does not actively search for it. Prefers this method because they are not good at checking events themselves, even though they would like to participate.

- **Person 5** Mainly uses social media such as Instagram because there are many different profiles sharing information.
- **Person 6** Receives information through friends or from notice boards near the large cafeteria.
- **Person 7** Uses Instagram and Facebook and also asks friends for information.

A.1.2 Question 2: What do you think about the quality of the information you receive about events? Are you ever unsure about where, how, or when events take place?

- **Person 1** Thinks the information is generally fine and usually includes dates and times. However, event descriptions are often lacking, and not all necessary information is always included.
- **Person 2** Believes the issue is not the quality of the information itself, but rather that it is not communicated effectively to students.
- **Person 3** Does not remember any major issues with the information provided.
- **Person 4** Finds the information clear and understandable. Feels well informed and has not experienced any problems.
- **Person 5** Is aware of the events but feels there is not enough information. Thinks the quality of communication is poor and that posts often feel rushed or incomplete.
- **Person 6** Is often unsure because there is too little information about the events themselves.
- **Person 7** Finds information difficult to locate because it is spread across multiple platforms. Once found, the information is usually acceptable.

A.1.3 Question 3: Do you think the information about social events meets your needs? Why or why not?

- **Person 1** Generally feels that the information is sufficient most of the time.

- **Person 2** Thinks the information is sufficient and likes being able to see how many people are attending. However, attendance indicators can also discourage participation if nobody signs up first.
- **Person 3** Feels the information is sufficient.
- **Person 4** Finds the information reliable and adequate.
- **Person 5** Feels that some events fail to capture interest because there is not enough information provided, such as who or how many people are attending.
- **Person 6** Thinks the information is sufficient to some extent, but lacks details such as attendance numbers.
- **Person 7** Finds the information somewhat superficial and would like more detailed descriptions of the events.

A.1.4 Question 4: When was the last time you attended a social event, and how was your experience finding information about it?

- **Person 1** Found the experience confusing because information was only shared through Messenger via a tutor. Had to call the tutor during the event to confirm the location and time.
- **Person 2** Attended a Christmas lunch event where information was shared on Facebook and Messenger, making students aware of the event.
- **Person 3** Attended a study start party. Information was shared through Facebook and Messenger, which worked well.
- **Person 4** Found the process easy and uncomplicated because all information was sent directly.
- **Person 5** Attended a bowling event. Information was shared through tutors and a Facebook group, making it easy to stay informed. Friends also helped by reminding them to participate.

- **Person 6** Attended a bowling event. Found it manageable because it was only for students from the same semester, and a group chat provided a clear overview of who was attending.
- **Person 7** Attended a Friday bar event two weeks ago and found information through Instagram.

A.1.5 Question 5: How do you feel about information regarding social events being spread across multiple platforms?

- **Person 1** Prefers having all information gathered in one place. Believes different platforms may provide inconsistent information and that a central platform would improve overview and accessibility.
- **Person 2** Thinks it is inconvenient both for organizers and students. Would like all information gathered on one platform and suggested notifications through student emails.
- **Person 3** Questions the purpose of Moodle Students if students still need to join Facebook groups. Suggested a schedule or calendar view with filtering options.
- **Person 4** Feels like the wrong target group because they do not actively use social media platforms. Is unsure whether they would actively seek information if necessary.
- **Person 5** Thinks multiple platforms can be beneficial for users who already use them, but posts are easy to forget. Would prefer if everything was collected in one place, such as Moodle.
- **Person 6** Finds multiple platforms confusing and would prefer a dedicated website or centralized platform. Believes this would increase interest in attending events. Also mentioned missing information in Messenger group chats because of excessive messages.
- **Person 7** Finds it frustrating to search for information across multiple platforms and believes it reduces motivation to participate in events.

A.2 Interviews

Person 1

Interviewer: How do you search for information about social events at Aalborg University CPH? Where do you usually find them?

Person 1: Normally, it is just on Facebook and Instagram from Aalborg University CPH Copenhagen. When they post there, that is actually the only place I look. Otherwise, I think it is really difficult to find the information myself.

Interviewer: So it is because you find it challenging to locate the information that you stick to one place?

Person 1: Exactly.

Interviewer: What do you think about the quality of the information you receive about the events?

Person 1: It is fine enough. But I do not know if it contains everything you should know. There is always some general information along with times and dates, so that is fine for me. But it is not always very detailed.

Interviewer: So you are missing a deeper description of what is actually going to happen at the event?

Person 1: Yes.

Interviewer: Do you think the information about the social events covers your needs?

Person 1: Yes, I think so most of the time.

Interviewer: When was the last time you attended a social event, and how

was your experience finding information about it?

Person 1: Good question. I actually cannot remember. Well, I did join the bar crawl. It was fine, but there was still no information other than what our tutor gave us on Messenger. We actually had to call her while we were on our way because we were not meeting at the school. We had to ask where it was because there was no information about the location or the meeting time. Everything was very mixed up.

Interviewer: So it was a confusing experience?

Person 1: Yes, it was a confusing experience.

Interviewer: How do you feel about the information regarding social events being spread across multiple platforms?

Person 1: You do not necessarily want to see the same thing on Instagram and Facebook. But if you knew there was one place where you could find all the information, then I think it would be much easier than having to search in several different places for the same information. Just one place where you know you can find everything you need.

Interviewer: Would that make you participate in more events?

Person 1: Yes, I would say so. Especially if there was more information about them. It is difficult if you do not know which events are happening.

Person 2

Interviewer: Where do you look for information about social events at Aalborg University CPH? And do you even look for it at all?

Person 2: No, I do not really look for it. But if I were to look for it, it would probably be on Facebook.

Interviewer: Mm.

Person 2: And honestly, I do not know where else you would check.

Interviewer: There are actually other places, for example Moodle Students. Did you know that existed?

Person 2: Yeah, I knew it existed.

Interviewer: Okay.

Person 2: But then you have to sign up for their Moodle Students thing. It is the groups feature, right?

Interviewer: Yes, you have to enroll yourself.

Person 2: Yeah.

Interviewer: This might be a bit off topic, but is there any reason why you do not look for social events?

Person 2: I do not know. I think it is just because everything is not gathered in one place.

Interviewer: Yeah.

Person 2: There is not really some kind of official school platform for it. Or maybe there is, but I do not know.

Person 2: Have you not heard about it?

Interviewer: No.

Person 2: Does the school not have something like CphCampus Events or something?

Interviewer: No. So it is more that you do not know who is responsible for what, or how big the events are?

Person 2: I do not even know if there are any events at all, honestly. Like, where would I even hear about an event if I do not know they exist?

Interviewer: If you do not know they exist?

Person 2: Exactly. I do not even know where I would go to look for events.

Interviewer: But you did mention Facebook.

Person 2: Yeah, but that is only because we are part of our study group with our coordinator or whatever her role is.

Interviewer: There are posters around campus and screens showing different things. Is it just because it does not appeal to you?

Person 2: I honestly just do not think I have noticed them. My thought is just that if I wanted to know whether something was happening, I do not think I would go look at some random bulletin board.

Person 2: I would probably just check the website, but I have not personally gone looking. So I do not even know if it exists.

Interviewer: So in reality, your need is that it should simply be communicated better that things are happening?

Person 2: Yes.

Interviewer: I do not know whether you have attended any events yet, but have you been to any events?

Person 2: Yeah, we went to one of the first ones. I cannot completely remember what it was.

Interviewer: Could it have been some kind of intro party?

Person 2: Yeah, I think it was an intro party. That is the only one I have

been to.

Interviewer: What was the quality of the information you received about it? Were you unsure about where to go, how it worked, or when it was happening?

Person 2: I honestly cannot remember. I do not think so.

Interviewer: No?

Person 2: It was early enough on, so yeah. I actually cannot remember exactly how it worked, but I do not feel like there were any issues.

Interviewer: So there was nothing wrong with the information itself?

Person 2: No.

Interviewer: It covered your needs sufficiently?

Person 2: Yeah, I think so. At least that is how I felt.

Interviewer: So overall, it is not communicated especially well?

Person 2: No. My thought was that it could be really nice if there was something similar to Moodle, where we have our schedules.

Interviewer: Yeah.

Person 2: The school does not necessarily need to use Moodle, but there could just be a calendar showing where all the events are.

Interviewer: So all events from the different committees would be collected in one calendar, where you could get an overview of things like Kaffepatten hosting something or Boulder having an event on Friday?

Person 2: That is what I am thinking.

Interviewer: How do you feel about information about social events being

spread across multiple platforms?

Person 2: I feel like you do not really know where things are. Let us say you go into Moodle and find the Student Moodle page, then you have to go onto Facebook and join their Facebook group. So what is even the point of Student Moodle then?

Interviewer: Regarding your idea with a calendar, could it become overwhelming if there were many different committees?

Person 2: Yeah, definitely.

Interviewer: Would it need some kind of filtering mechanism?

Person 2: My thought is that if I am part of the Boulder community, then it would be nice to get all their updates and see what they are doing and on which days. But if I am not part of that community, then it is maybe not that important for me to see it.

Interviewer: So you would like the option to filter things out, while still being aware that these communities exist?

Person 2: Yeah. Maybe you could join groups on the platform itself and then receive updates only from the ones you choose to follow. Then maybe there could also be an overall section where you can see what Aalborg University CPH itself is organizing.

Interviewer: But could that mean people might miss out on cool events organized by others?

Person 2: Maybe. I do not know. If you are already part of a club, then maybe their regular weekly activities are not that important to have on a big calendar. But if they are hosting some kind of special or major event, then maybe it should appear there.

Interviewer: A larger event or something like that?

Person 2: Exactly. I do not know, it was just an idea.

Person 3

Interviewer: Valdemar, where do you look for information about social events at Aalborg University CPH?

Person 3: Either through messages in the Messenger group, or if it gets posted in the Facebook group.

Interviewer: Is that something you actively keep up with?

Person 3: Not really. I get notifications from Messenger, but not from Facebook. Usually, I only find out about events if someone mentions them.

Interviewer: So it is not really something you actively search for yourself, you mostly get told about it?

Person 3: Yeah. It would be nice if there was an easier way to find out about things. Like, if there was one place where everything was easy to find and organized.

Interviewer: What do you think about the quality of the information you receive about events? Are you ever unsure about when things happen or how they work?

Person 3: Most of the time, when events are first announced, it is difficult to figure out what is actually going to happen. It is usually only about a week or a few days before the event that you can finally understand what the event is really about.

Interviewer: Is that a major problem for you, or is it manageable?

Person 3: You can live with it, but it would definitely be better if there was more information earlier.

Interviewer: So you do not really feel that the information fully covers your needs? You would like more detailed information?

Person 3: Yeah.

Interviewer: When was the last time you attended a social event, and how was your experience finding information about it?

Person 3: I think it was the Christmas party, and there was a Facebook event where I could find the information.

Interviewer: Have there been other events you considered attending after seeing information about them, or was it more that you just were not interested?

Person 3: No, there were other events I wanted to attend, but I had to work, so I could not go.

Interviewer: So planning is a problem?

Person 3: Yeah. In terms of getting time off work, information about events comes out way too late.

Interviewer: So the information is simply communicated too late?

Person 3: Yeah, it is not announced early enough for me to request time off work.

Interviewer: How do you feel about information about social events being spread across multiple platforms?

Person 3: In one way, it is nice that information is shared in multiple places because not everyone uses the same platforms. But at the same time, it is really annoying that everything is not gathered in one place, because sometimes different information appears in different places.

Interviewer: Does that also make the information more confusing when different things are written in the Facebook event, Messenger group, Discord, and so on?

Person 3: Yeah. For example, in a Facebook event, people write the time both as part of the event details and again in the description, and sometimes the times do not match. That is just really frustrating.

Person 4

Interviewer: Where do you look for information about social events at Aalborg University CPH?

Person 4: Um, I usually get the information sent directly to me.

Interviewer: So you do not actively search for it yourself?

Person 4: No.

Interviewer: So that is the only way you hear about social events?

Person 4: That is the only way I hear about social events. Honestly, I am really bad at checking things myself. I usually check my own schedules and tools the evening before I have to go to work.

Interviewer: Is that because you are not interested in participating in events?

Person 4: No, I do want to participate in things. I am just not very good at checking for information myself.

Interviewer: What do you think about the quality of the information you receive about events? Are you ever unsure about where, how, or when things happen?

Person 4: No.

Interviewer: Could you elaborate on that?

Person 4: Well, I usually get messages explaining what is happening and when it is happening. So I feel like I generally have a pretty good understanding of what is going on and when.

Interviewer: Do you think the information you receive is detailed enough?

Person 4: Yes. I have not had any problems so far, honestly.

Interviewer: When was the last time you attended a social event, and how was your experience finding information about it?

Person 4: Let me think. The last social event I attended was probably the Christmas party. It was pretty easy. We received all the information directly, so there was nothing particularly difficult about it.

Interviewer: How do you feel about information about social events being spread across multiple platforms?

Person 4: I am probably the wrong target group for that question, because I do not really use those platforms very much.

Interviewer: What if you personally had to actively search for information yourself?

Person 4: I do not know. It is difficult to force people to actively search for information themselves if they do not already want to.

Interviewer: So you would never actively search for information about an event yourself if you were going to attend it?

Person 4: I do not know. I have not really done that before. I usually just ask other people.

Person 5

Interviewer: Where do you look for information about social events at Aalborg University CPH?

Person 5: Primarily on social media such as Instagram, because they have many different profiles. Sometimes I just check what they have posted, and that is mainly where I get information from.

Interviewer: What do you think about the quality of the information you receive about the events? Are you ever unsure about where, how, or when things happen?

Person 5: Yeah, the only useful thing is that I know an event exists, but I often do not know when it is happening. I do not think the quality of the information is as good as it should be. It almost feels like they just post something without giving much actual information about it.

Interviewer: Do you think the information about the social events covers your needs? Why or why not?

Person 5: Not really. To be honest, I do not think the events really match my interests. They have some good events for people who are maybe more "nerdy" or interested in specific things, but otherwise I do not feel like the events really appeal to me. I also do not feel like I get enough information about them. If I do not get enough information, I quickly lose interest. You just see a quick post, but you are told almost nothing about who is coming, how many people will attend, or what the event is really about, and then you just lose motivation to go.

Interviewer: When was the last time you attended a social event, and how was your experience finding information about it?

Person 5: I attended a bowling event, and I actually thought that was really good. We received a lot of information, both from our tutors who organized the event. Our tutor Katrine gave us a lot of information and kept reminding us about it, which made it easier to participate. I received the information through Facebook, where there is a Facebook group for software students. It also helps a lot when your friends keep reminding you to attend, and when people repeatedly share all the details, such as times, locations, addresses, and other information across the groups.

Interviewer: How do you feel about information about social events being spread across multiple platforms?

Person 5: On one hand, I think it is good because people use different platforms, whether that is Instagram, Facebook, or TikTok. So it makes sense that the information is spread across different platforms. But at the same time, I think it could be easier if everything was gathered in one place that you could quickly check. For example, Moodle. Then you could just check Moodle and see whether any events are happening. I feel like that would make it easier to find different events. Sometimes when you look at Instagram, you quickly forget the post you saw. So if everything was collected in one place, I think everyone would have a better chance of seeing it, and it would be easier and more effective.

Person 6

Interviewer: Where do you look for information about social events at Aalborg University CPH?

Person 6: If it is not something I hear about through my friends and it is something I want to search for myself, then I know there is a bulletin board near the large cafeteria, but I rarely go there.

Interviewer: What do you think about the quality of the information you receive about events? Are you ever unsure about where, how, or when things happen?

Person 6: Most of the time, yes. I think there is too little information about the actual event itself.

Interviewer: Do you think the information about the social events covers your needs? Why or why not?

Person 6: To a certain extent. You usually know approximately when the event is taking place, but again, if something changes, I do not think enough information is shared about it, or about how many people are attending and things like that. I just do not think there is enough information.

Interviewer: When was the last time you attended a social event, and how was your experience finding information about it?

Person 6: The last time was when we went bowling with our semester group. I thought that experience was better because it was not an event open to the entire school, so it was easier to keep track of things like who was attending and where it was happening, since we had a dedicated group for the semester.

Interviewer: How do you feel about information about social events being spread across multiple platforms?

Person 6: It supports the idea that things become confusing, because one thing might be written on Facebook, while something different appears on posters around campus or on Instagram. I would prefer if there was just one website or one place where everything was collected, so you could be sure that the information there was the correct version. I would also be more interested in participating in events if there was more information available, because then you would know what you were getting into. I also feel like you do not

hear much about many of the events. A lot of people spam messages in our Messenger group, but I do not often read all the messages.

Person 7

Interviewer: First question: where do you look for information about social events at Aalborg University CPH?

Person 7: It varies a bit. Either on Instagram or Facebook. Sometimes I also ask people. It is kind of trial and error until you find what you are looking for.

Interviewer: So you ask your friends on campus?

Person 7: Yes.

Interviewer: What do you think about the quality of the information you receive about events? Are you ever unsure about where they are or when they take place?

Person 7: I think it is very unclear where you are actually supposed to find the information. Once you find it, the information itself is usually okay, but you have to search around on different platforms.

Interviewer: So you think it is a problem that the information is spread across different places?

Person 7: Yes, it is difficult to find.

Interviewer: Do you think the information about social events covers your needs?

Person 7: I think there could definitely be more information. Sometimes it feels a bit superficial, so it would be good to have more details.

Interviewer: So there is too little information?

Person 7: Yes.

Interviewer: When was the last time you attended a social event, and how was your experience finding information about it?

Person 7: It was the Friday bar two weeks ago, I think.

Interviewer: Was it difficult to find information about the event?

Person 7: That one was actually okay.

Interviewer: So you did not have any major problems with it?

Person 7: No.

Interviewer: Where did you find the information?

Person 7: On Instagram.

Interviewer: Last question: how do you feel about information about social events being spread across multiple platforms instead of being gathered in one place?

Person 7: It makes things frustrating, and it makes you less aware of what is happening. It also makes you less motivated to participate because you do not want to spend time searching around for all the different pieces of information.

Interviewer: Okay. Thank you for your help.

A.3 Affinity Diagram

Interview question	Findings from the affinity analysis
Question 1: How students search for information	• Social media • Public displays such as posters and notice boards • Verbal communication from friends • Some students do not actively search for information
Question 2: Quality of the information	• Some students experienced no major issues • Information was sometimes incomplete or unfinished • Information could be difficult to find
Question 3: Whether the information was sufficient	• Some students found the information sufficient • Lack of information regarding number of participants • Difficult to maintain an overview • Information could appear superficial • Some students lost interest
Question 4: Last event attended	• Bowling – good information • Study start party – good information • Christmas lunch – good information • Friday bar – good information • Missing information about other events
Question 5: Whether the information was scattered	• Preference for having all information gathered in one place • Information was experienced as frustrating and confusing • Event organizers experienced similar challenges • Lower motivation to participate • Students forgot posts about upcoming events

Table A.1: Overview of themes identified in the affinity diagram analysis.

A.4 Unit testing code

A.5 Code example of the test related events

eventTest.js :

```
import request from "supertest";
import app from "../server.js";

describe("GET /events", () => {
```

```
5
6       test("should retrieve event data correctly", async () => {
7
8           const response = await request(app).get("/events");
9
10          console.log("Returned event data:");
11          console.log(response.body);
12
13          expect(response.statusCode).toBe(200);
14
15          expect(Array.isArray(response.body)).toBe(true);
16
17          expect(response.body.length).toBeGreaterThan(0);
18
19          expect(response.body[0]).toHaveProperty("title");
20          expect(response.body[0]).toHaveProperty("date");
21          expect(response.body[0]).toHaveProperty("location");
22
23      });
24
25  });
26
27  describe("PATCH /events/:id", () => {
28
29      test("should update event data correctly", async () => {
30
31          const updatedEvent = {
32              title: "Updated Test Event",
33              date: "2026-05-20",
34              timeStart: "12:00",
35              timeEnd: "14:00",
36              location: "AAU Test Room",
37              description: "Updated through unit testing",
38              practicalInfo: "Bring laptop"
39          };
40
41          const response = await request(app)
42              .patch("/events/1")
43              .send(updatedEvent);
44
45          console.log("Updated event:");
46          console.log(response.body);
47
48          expect(response.statusCode).toBe(200);
49
50          expect(response.body.title).toBe("Updated Test Event");
51
```

```
52      });
53
54  });
```